

Bidirectional Electrothermal Inchworm Actuator

Group E

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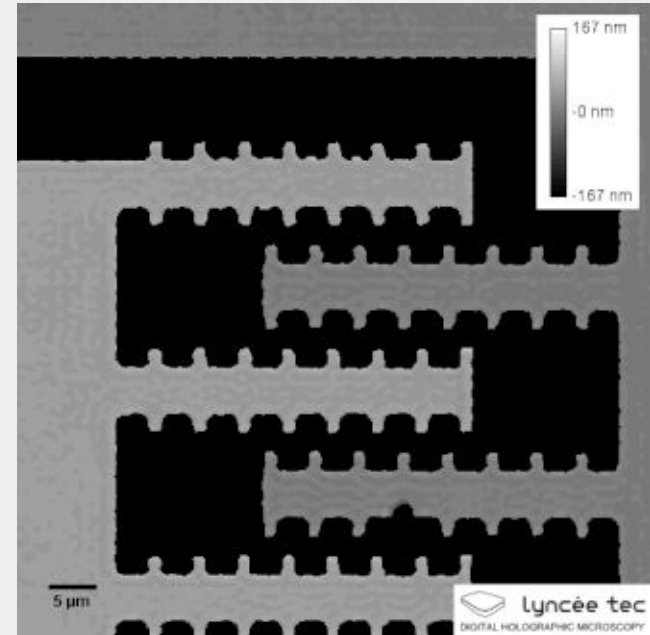
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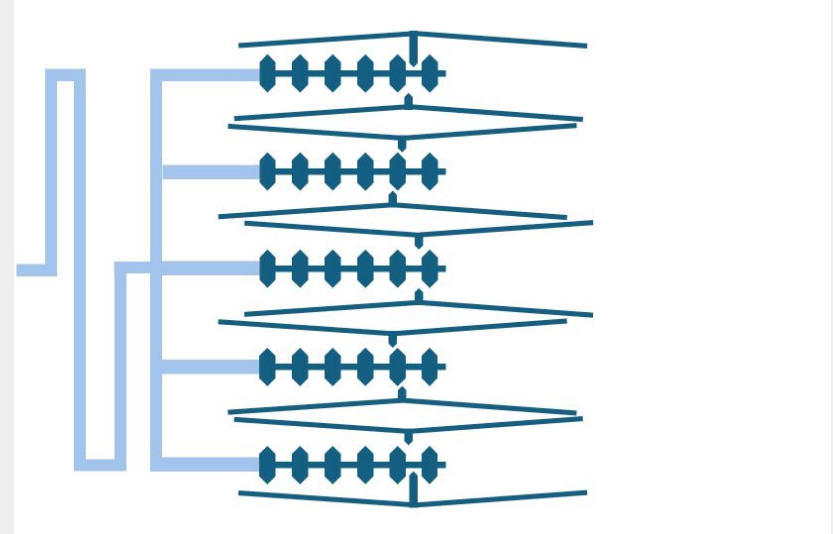
Project Motivation

- MEMS actuators are limited to small displacements - many basic designs (electrothermal & electrostatic) within our size constraints have maximum displacements ~15-20 microns [1]
- Sustained displacements require lots of power, especially electrothermal [2]
- We propose a solution to achieve very high displacements and sustain them for long periods of time with little to no steady state power usage



Operation Principle

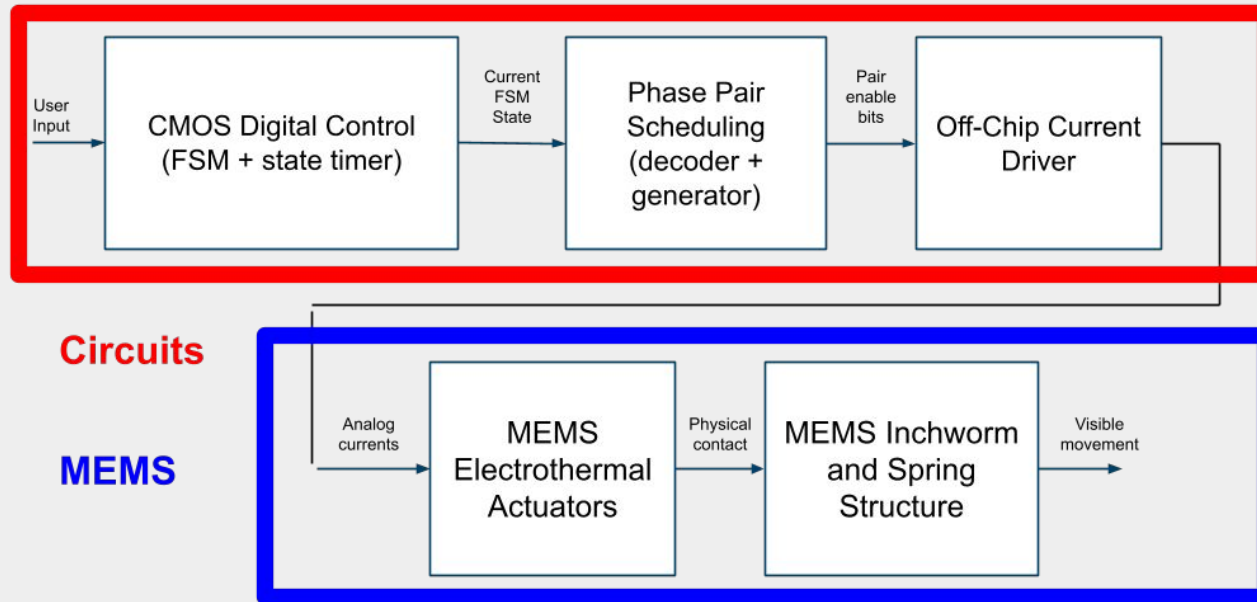
- Achieve high displacements with multi-stage actuation technique shown on the right
- Utilize a series of actuator pairs to “pinch” and move a large central structure attached to a spring
- Operate actuator pairs in reverse to generate movement in the other direction
- Outer actuators have reverse polarity to save position when device is turned off



****NOT OUR OFFICIAL MEMS LAYOUT**

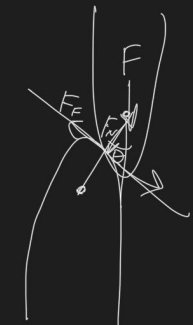
This illustrates the principle of movement for our device

Overall Microsystem Operation



Goals

- Prove viability of general concept to enable design of more complex devices
 - Show that thermal and electrical conduction issues at points of physical MEMS contacts can be overcome
 - Verify that force generated from electrothermal actuation can be sufficient to move a central proof mass on a spring
 - Learn more about mechanical contact physics at a microscopic level
 - Friction, compression, wear



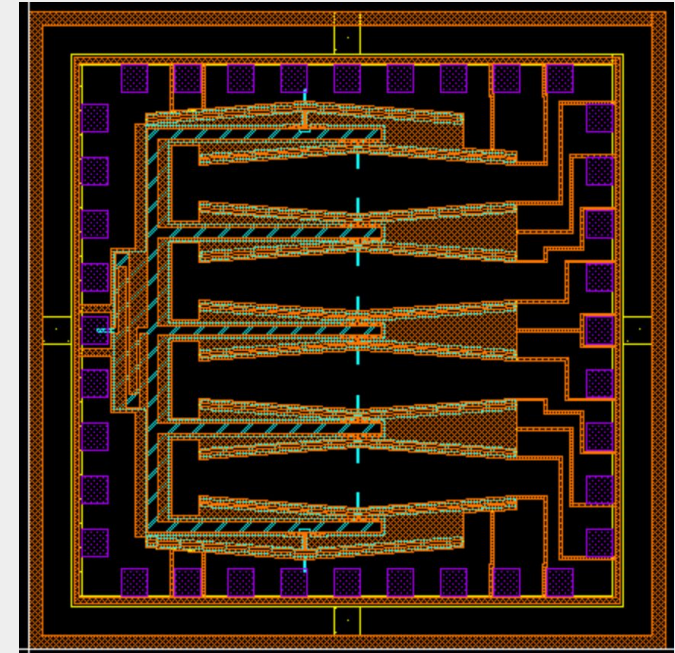
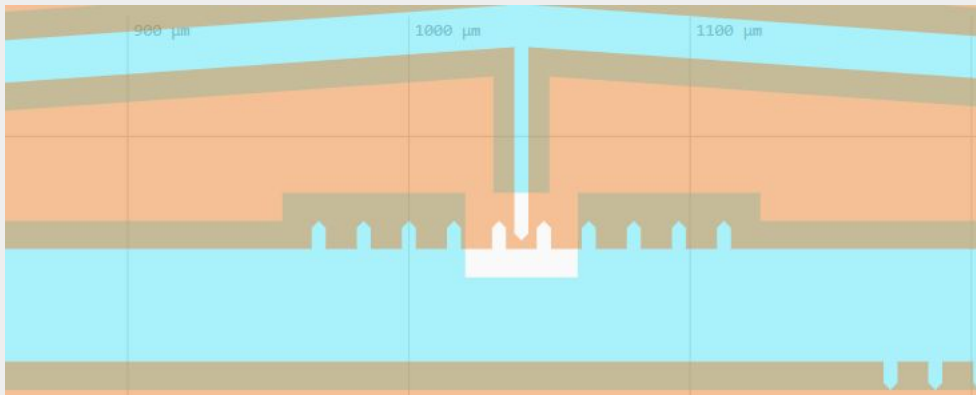
2 mN
 $F_N = F \cos \theta = 1.866 \text{ mN}$
 $F_F = F_N \mu = 1.866 \text{ mN} \cdot \mu$
 $F_s = F \sin \theta = 0.72$
 $0.72 = 1.866 \cdot \mu$
 If $\mu < 0.386$ goal
 $\theta = 21^\circ$

We require a coefficient of friction at the contact < 0.386 in order to overcome static friction, but there isn't much data on what to expect here

MEMS

MEMS Overview

- Our device separates these “stages” into vertical levels, as our actuators take up lots of horizontal space
- Outer stages operate in reverse polarity - preserves states when turned off

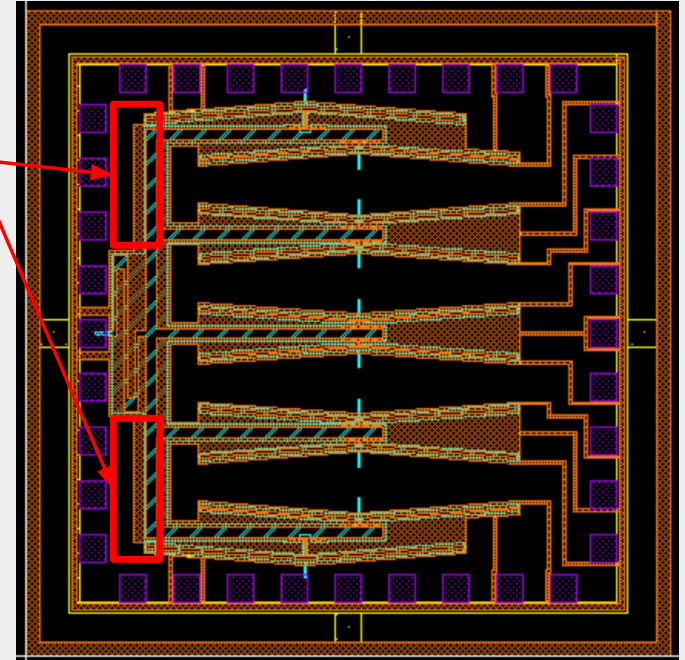


Orange - DRIE
Blue - Platinum
Black - Anchored Silicon
Yellow - Glass Recess
Purple - Aluminum

Potential Applications

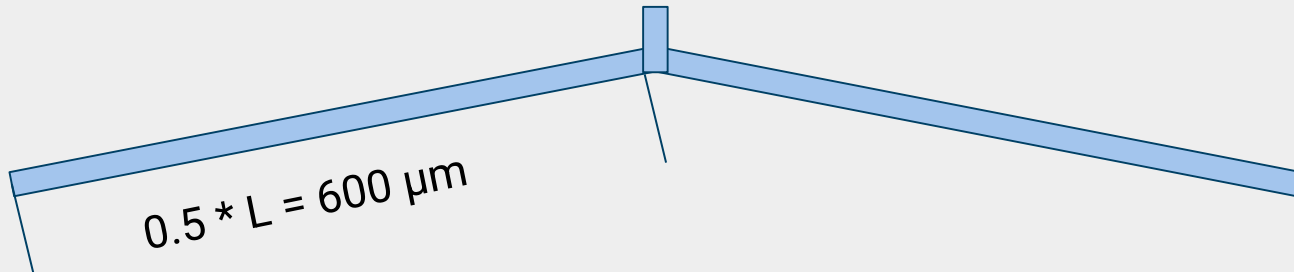
- Our device doesn't have an explicit onboard application, but is rather a proof of concept for a new method of actuation
- Some future designs enabled by our research include:
 - Varying gap/overlap capacitor with discrete states
 - Toggleable optical shutter
 - Microrobot impulse generator for locomotion

Theoretical
Capacitor
position



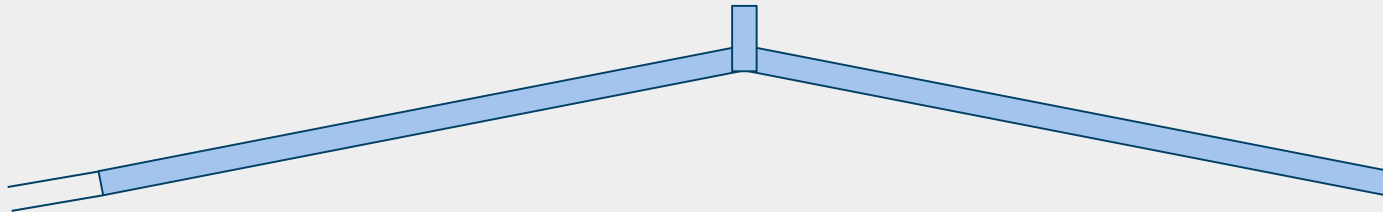
Actuator Length

- Length has the largest effect on displacement, so we maximized it to fit our area requirements
- Size is limited due to 2mm*2mm design constraint
 - 600 micron beams, or 1200 microns for the full actuator, was the maximum realizable size before cutting into space used for power lines and target movement specs



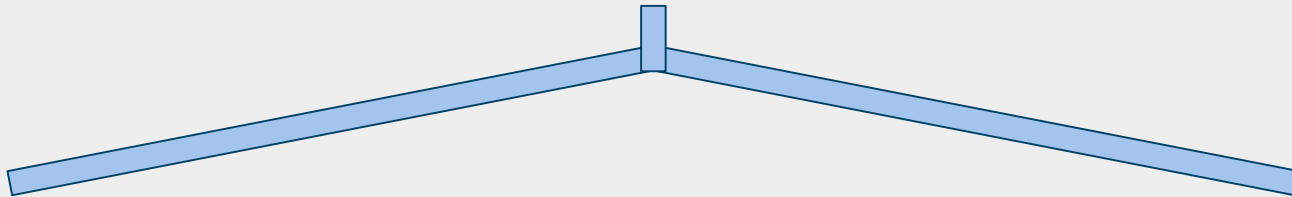
Actuator Width

- Width has a smaller effect on displacement, but we still tried to optimize it (keep it small)
- Too small of a width decreased output force and increased buckling propensity, also DRIE becomes riskier (1mm beam requirement)
- Too large of a width increases power requirements and decreases displacement



Actuator Angle

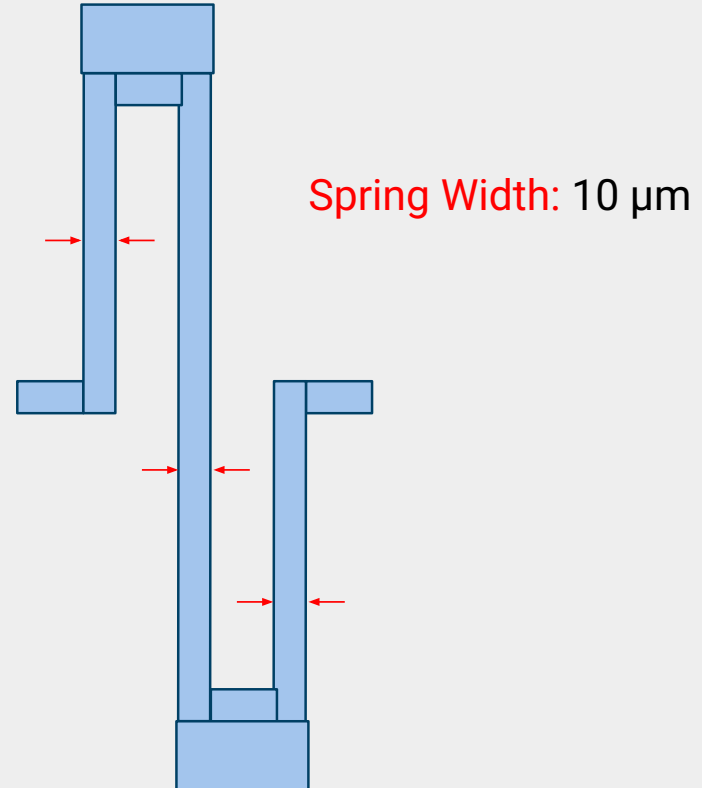
- The angle for a single stage bent-beam actuator has a trade-off between force produced and displacement
- To move the spring at the boundaries (near 50 microns) we need to produce at least 12.2 mN, requiring an angle of 4 degrees





Spring Width

- Spring width minimized to allow for a small spring constant while fitting the 1 mm design guideline
- To help with fabrication, width is 2 times larger than minimum design constraint and large “heat-sink” blocks were added to help with thermal dissipation



Simulations vs Calculations

Parameter	Simulation	Calculation
Actuator Displacement	8 microns	8.649 microns
Actuator Force	29 mN	15.27 mN
Actuator Current	70 mA	48.8 mA
Thermal Dispersion at Contact (max temperature drop)	-168° C	-202° C

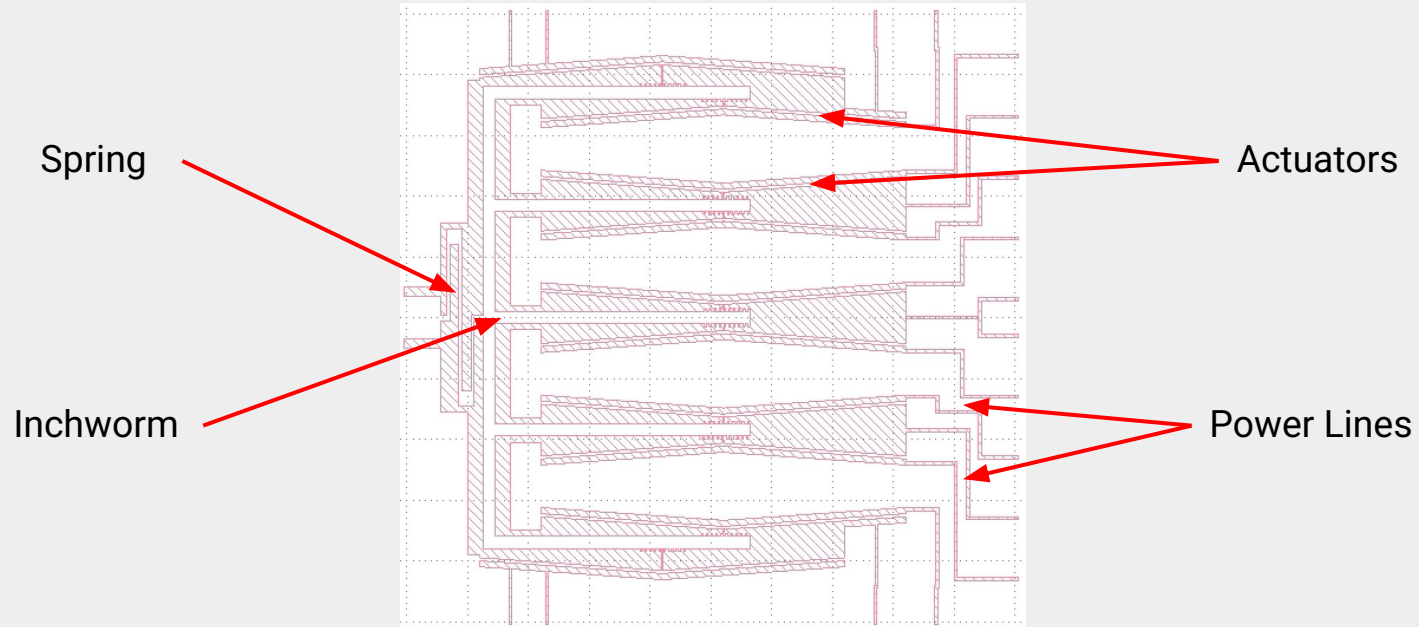
*Specifics shown in Appendix

Simulations vs Calculations

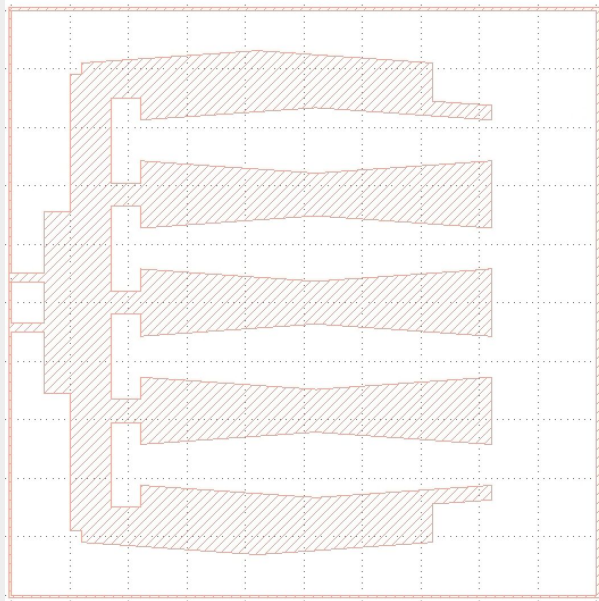
Parameter	Simulation	Calculation
Inchworm Lateral Spring Constant	87.8 N/m	96 N/m
Inchworm Vertical Displacement	0.05 microns	0.155 microns
Inchworm Squeeze Spring Constant	650 N/m	644.741 N/m
Buckling Force	18 mN	17.276 mN

*Specifics shown in Appendix

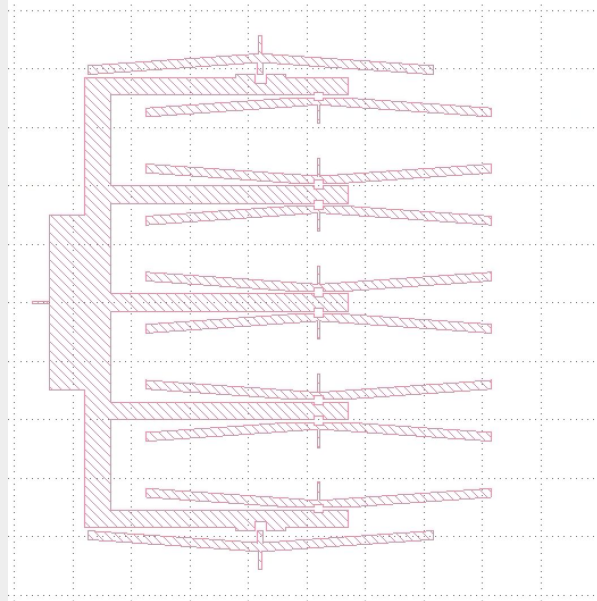
DRIE Layer



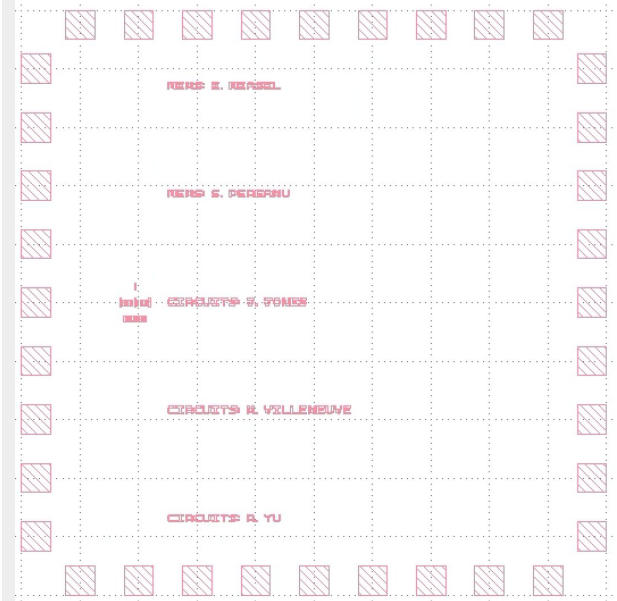
RECESS



METAL



PAD

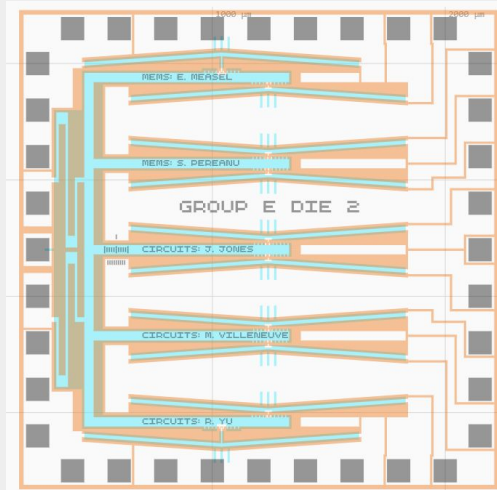


MEMS Specifications

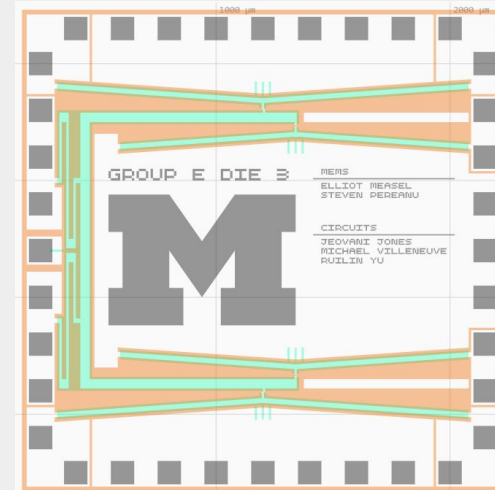
Specification	Anticipated Results
Target Minimum Inchworm Displacement	+/- 13 microns lateral movement
Maximum Displacement	+/- 50 microns lateral movement
Displacement Resolution (powered)	3.2 microns
Displacement Resolution (unpowered)	16 microns
Actuator Current	95 mA
Max Power Required	3-6 W
Temperature Reached	400° C

Additional Dies

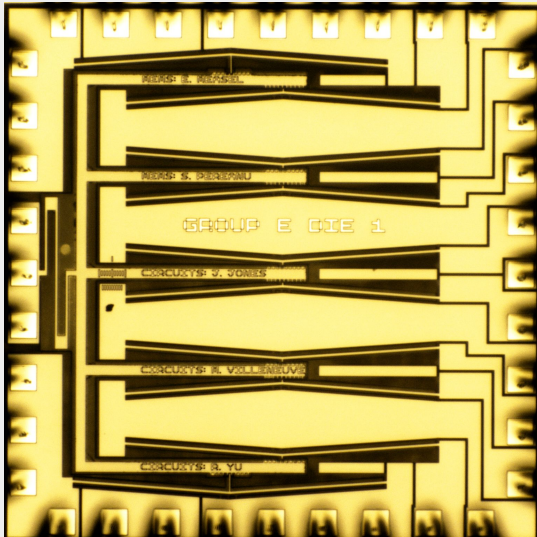
Die 2: 45° sharpened teeth, thicker actuators (15 μm), and extra spring



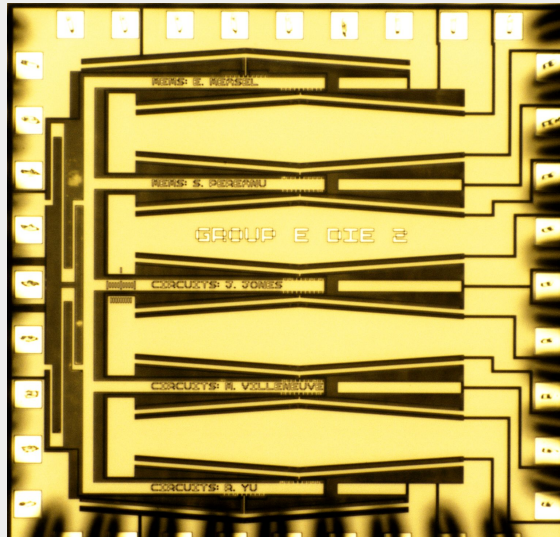
Die 3: only two stages and even thicker actuators (20 μm)



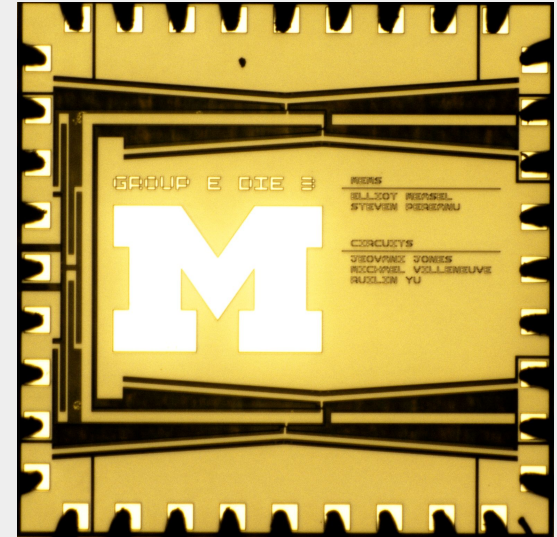
Fabricated Devices



Die 1: 10 μm actuators,
rounded teeth, 1 spring



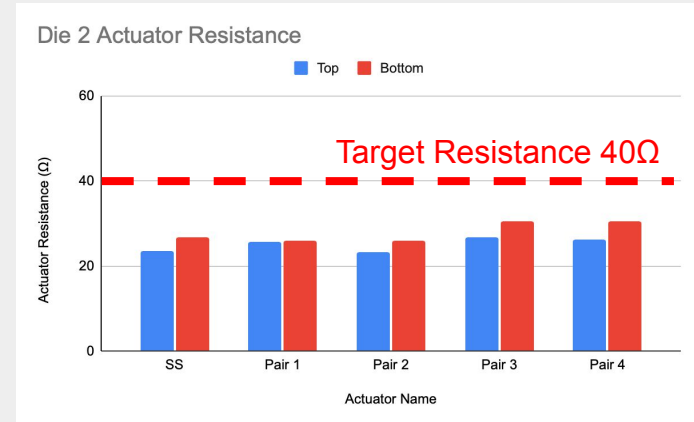
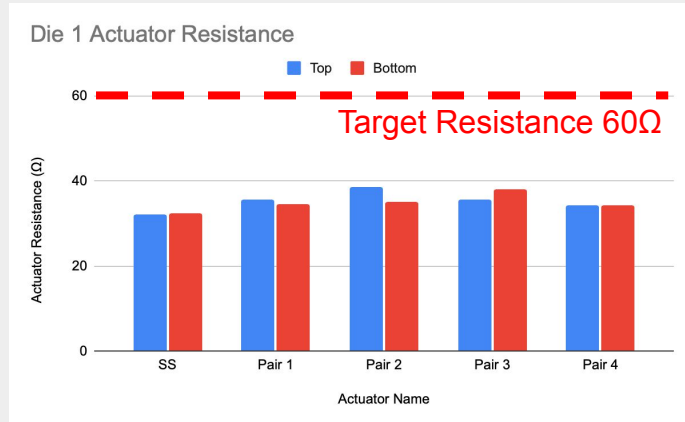
Die 2: 15 μm actuators,
sharp teeth, 2 springs



Die 3: 20 μm actuators,
2 stage design, 2 springs

MEMS Actuator Resistances

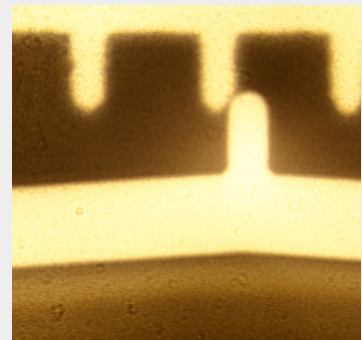
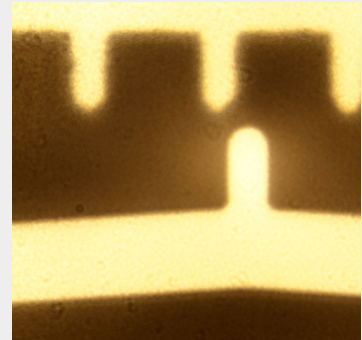
- Measured resistance across actuators was consistently lower than calculations
- This raised concerns on whether we would be able to supply enough power to get our desired displacement



Values obtained above were measured from dies near the edge of the wafer to try and get higher resistances - “best case values”

MEMS Die 1 Testing Results

- Arms give sufficient displacement ($8\text{ }\mu\text{m}$) with 95 mA input
- No buckling observed, even with physical contact
- Die 1 arms struggled to push the inchworm to the side, instead bending the inchworm arm upwards - potential issues include:
 - Friction at interface is too high to initiate sliding
 - Lateral spring constant is too high
 - Vertical spring constant of inchworm is too low to allow left-right movement
 - Resistance asymmetry prevents vertical force cancellations, doesn't redirect force into x-direction
 - Debris left over from fabrication could add adhesion or increased stiction that wasn't accounted for



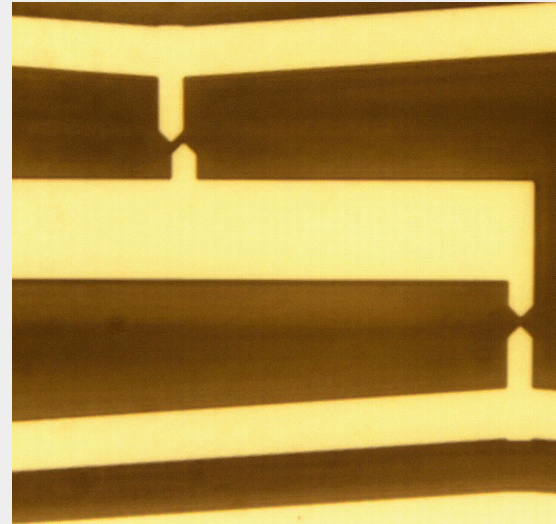
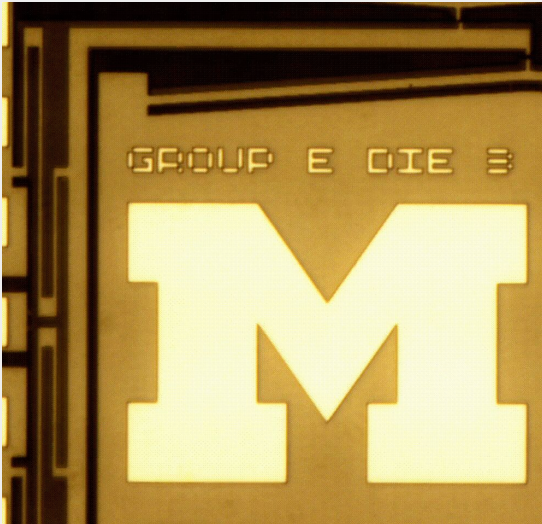
MEMS Die 2 Testing Results

- Most of the time, we observed the same issues on die 2 as we did on die 1
- One test showed promising results, moving the inchworm by ~ 3 microns using physical contact from the second stage
- Power was more of an issue here, as actuator resistances were much lower
 - Current limit was quickly reached



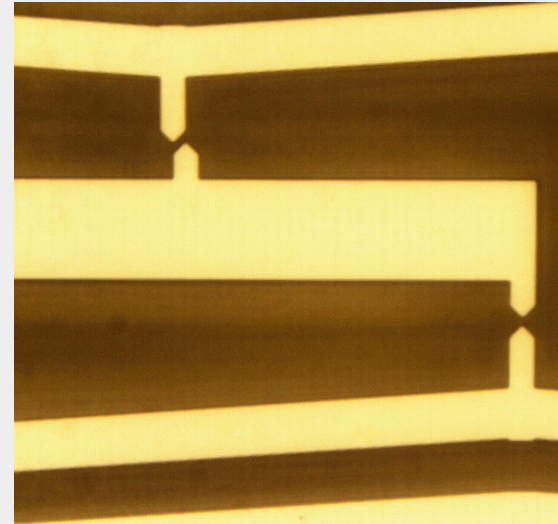
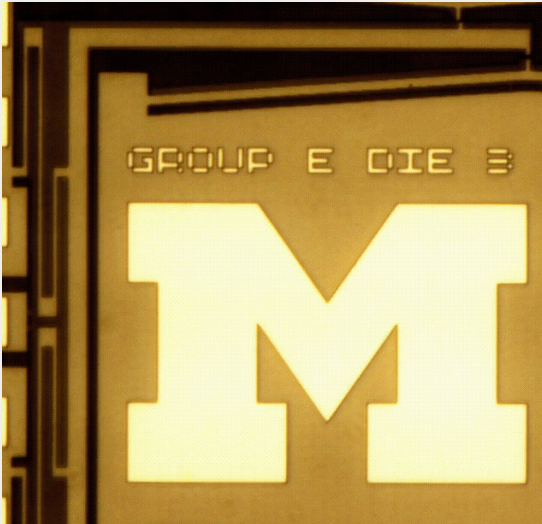
MEMS Die 3 Testing Results

- Die 3 functions as planned, proving our idea feasible
- First and second stage both engage with inchworm to translate vertical motion into lateral displacement



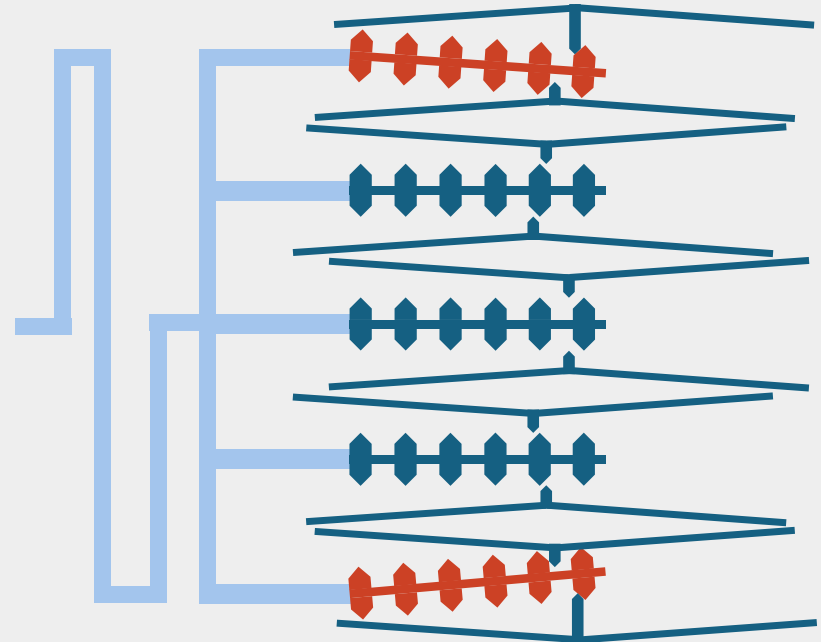
MEMS Die 3 Testing Results

-



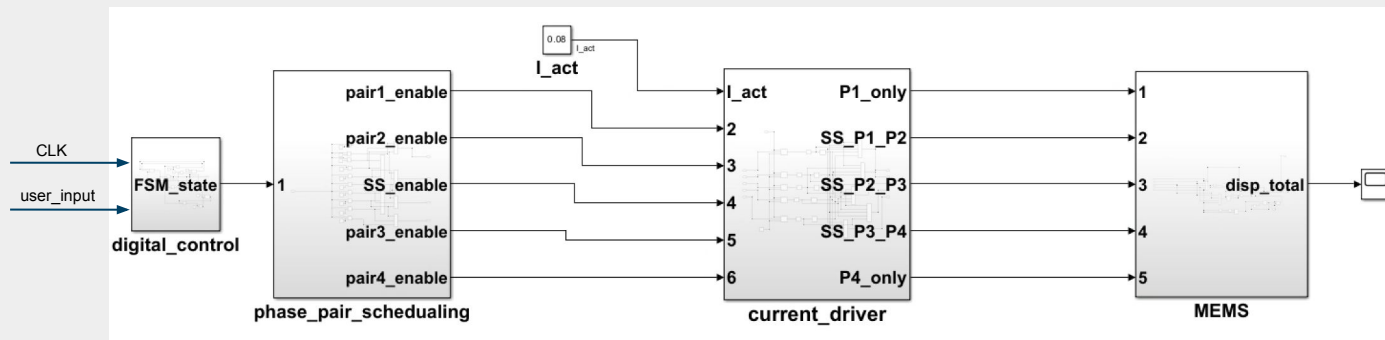
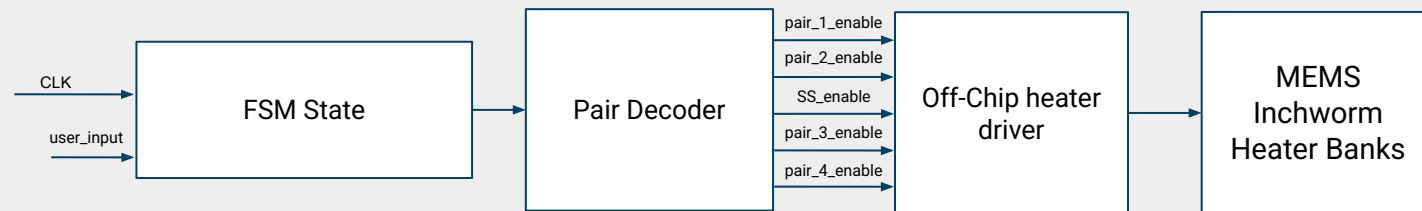
MEMS Proposed Changes

- Thicken inchworm squeeze spring constant
 - Too much compression at contact
 - This is a result of high friction coefficient as calculated for - sliding is hard to initiate

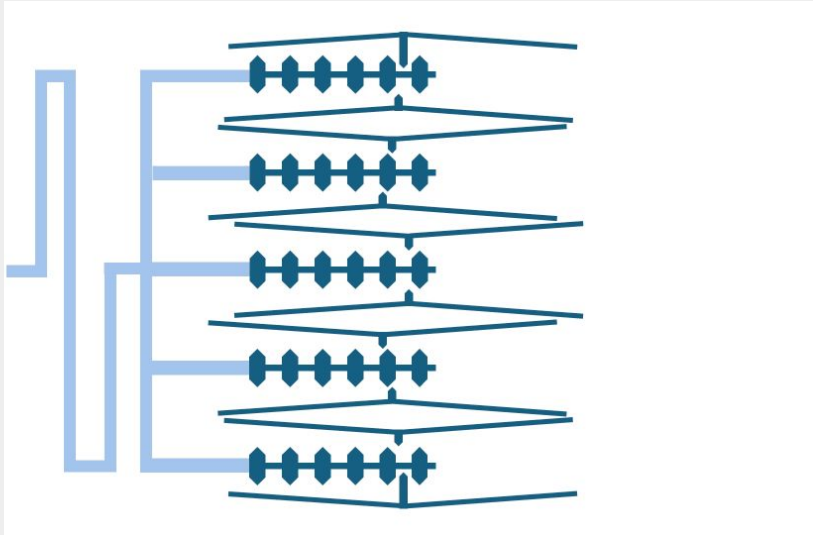


Circuits

Circuit Block Diagram



Top-level Circuit Operation

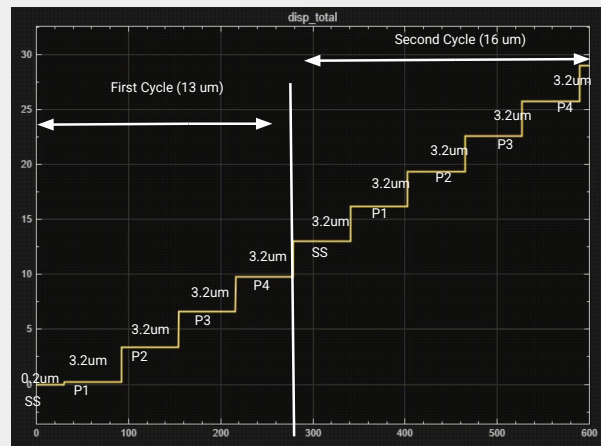


Yellow Acuator: get 100mA current and pinch



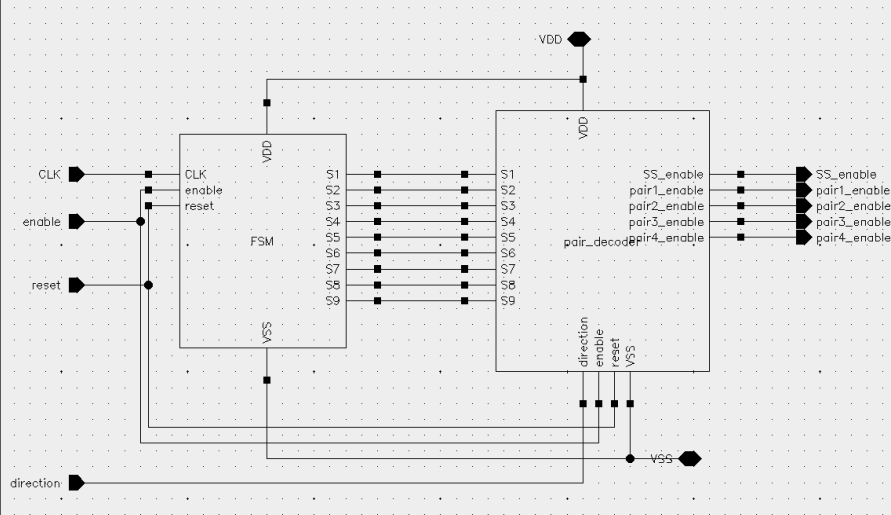
Logic phase sequence

Top-level Circuit Operation



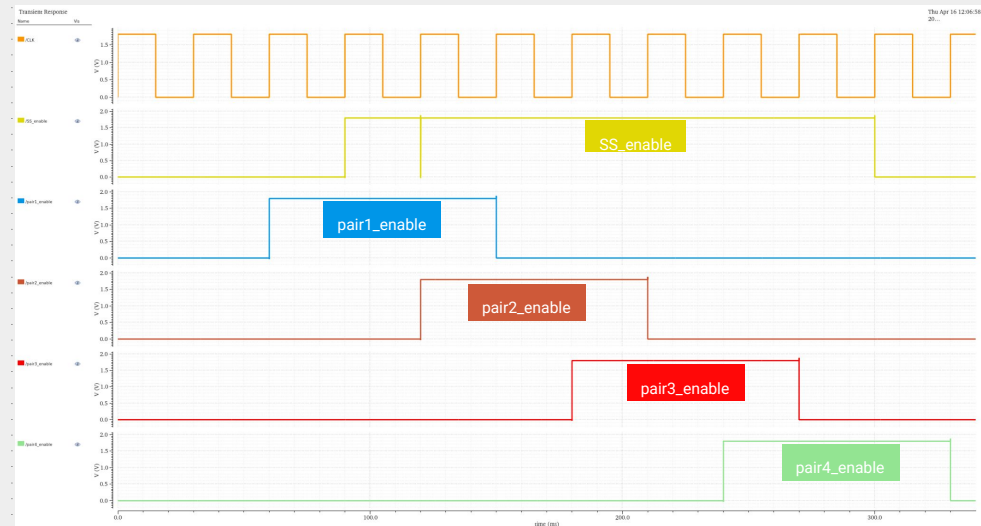
Corresponding motion per cycle

Top-level System Schematic & Simulation



Top-level schematic

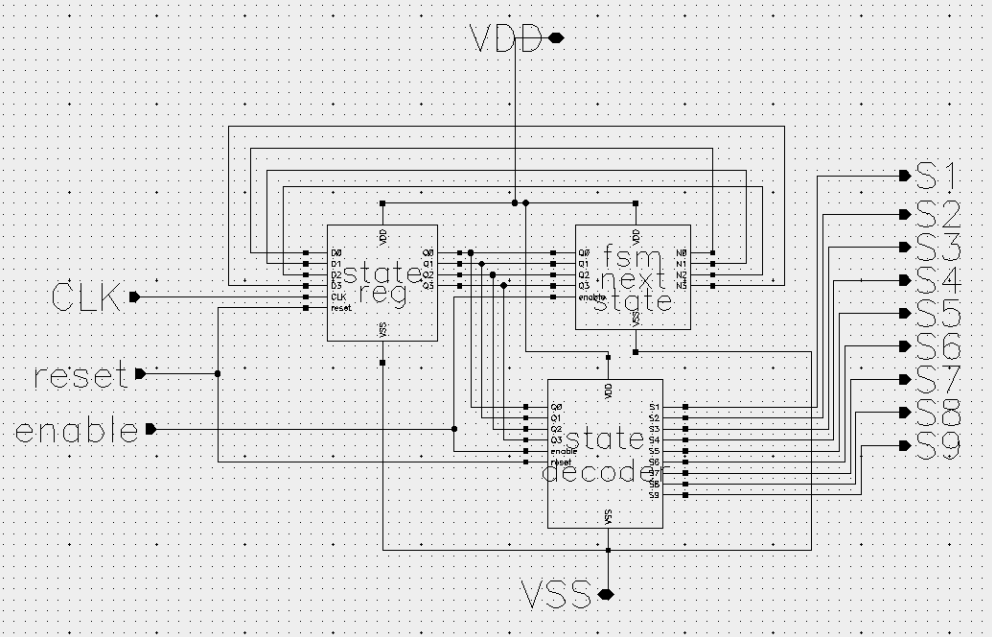
Top-level schematic is consist of FSM and pair_decoder, with inputs of CLK, enable, reset and direction, outputs of SS_enable, pair1_enable - pair4_enable



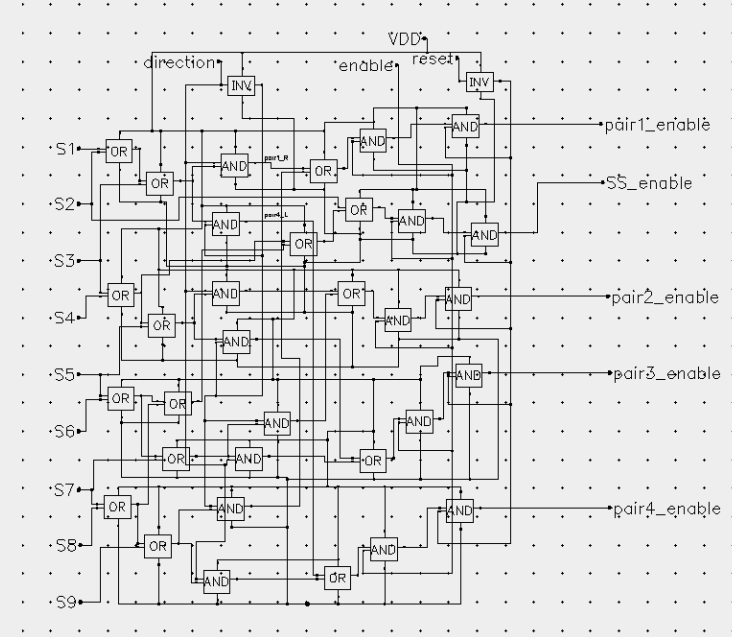
Top-level simulation

With 33 Hz CLK input, the system can generate SS_enable, pair1_enable - pair4_enable with overlap between each other, making the corresponding actuator move

Digital Control Schematics



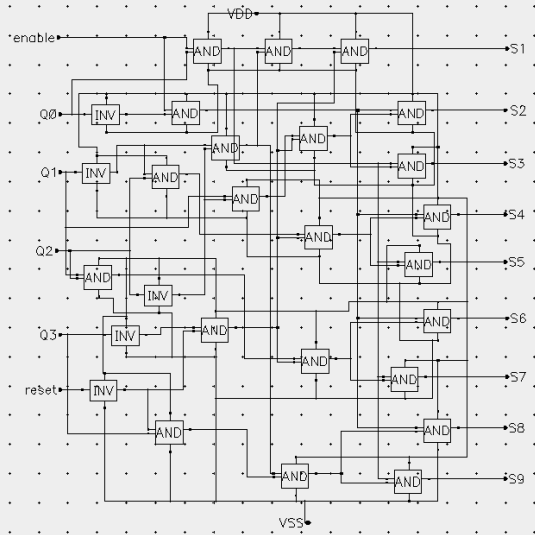
FSM



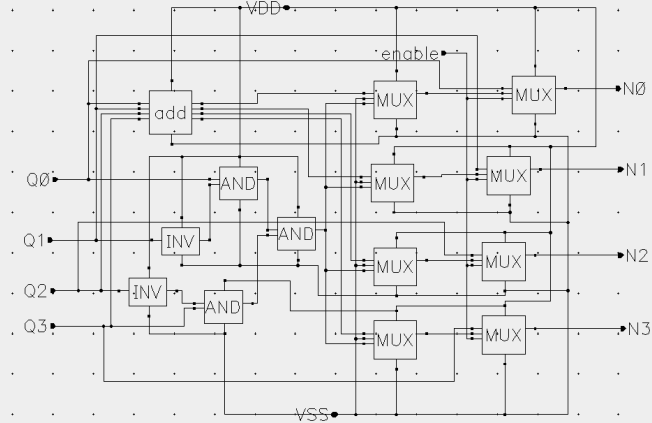
Pair Decoder



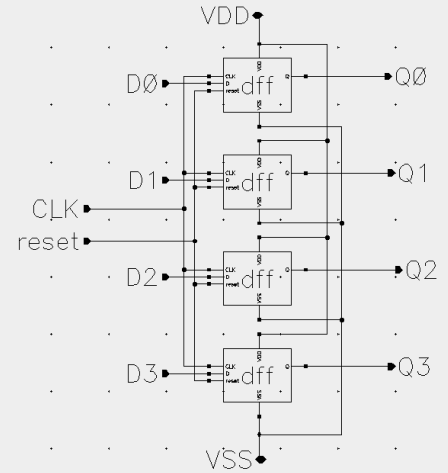
Digital Control Schematics



state_decoder



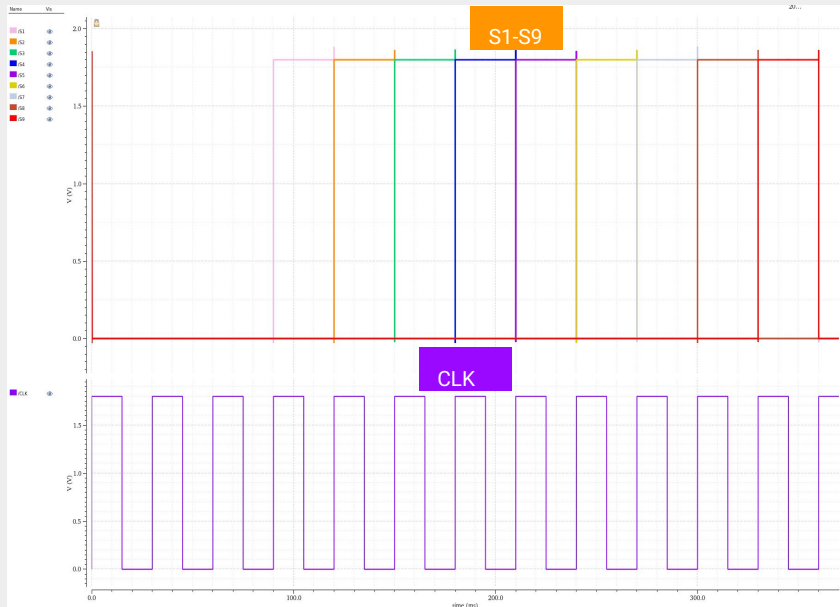
fsm_next_state



state_reg



Digital Control Simulation



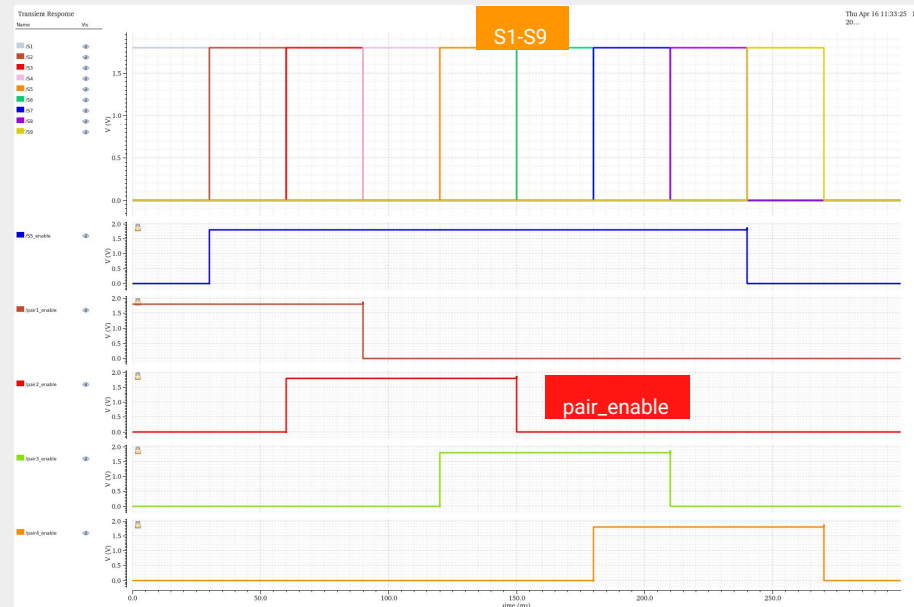
FSM

Input: CLK

output: S1-S9

Main Path:

CLK → S1-S9 → pair_enable → off-chip current driver → MEMS

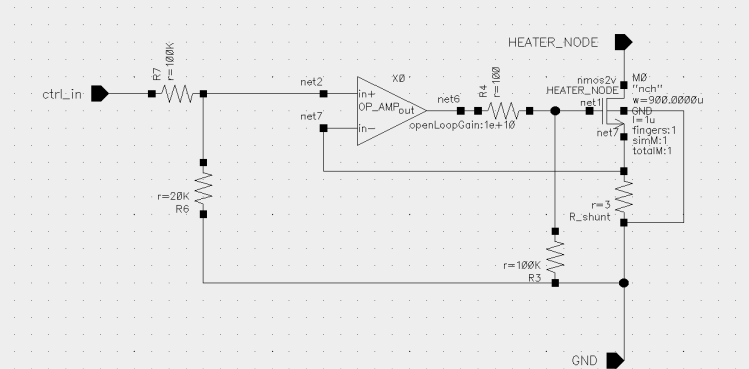


Pair Decoder

Input: S1-S9

output: pair_enable

Current Driver



$$V_{fb} = I_{out} R_{shunt}$$

$$V_{ref} = V_{fb}$$

$$I_{out} = \frac{V_{ref}}{R_{shunt}}$$

$$V_{ref} = V_{ctrl-in} \frac{R_6}{R_6 + R_7}$$

$$I_{out} = 100 \text{ mA}, \quad V_{ctrl-in} = 1.8 \text{ V}$$

$$V_{ref} = 0.3 \text{ V}$$

$$R_{shunt} = \frac{V_{ref}}{I_{out}} = \frac{0.3}{0.1} = 3 \Omega$$

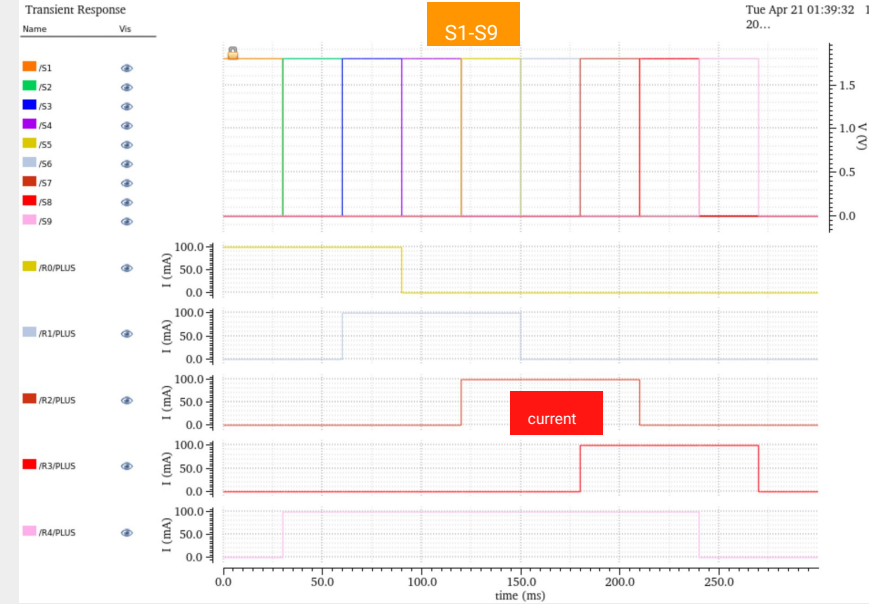
$$\frac{R_6}{R_6 + R_7} = \frac{V_{ref}}{V_{ctrl-in}} = \frac{0.3}{1.8} = \frac{1}{6}$$

$$R_6 = 20 \text{ k}\Omega, \quad R_7 = 100 \text{ k}\Omega$$

$$V_{ref} = 1.8 \times \frac{20}{20 + 100} = 0.3 \text{ V}$$

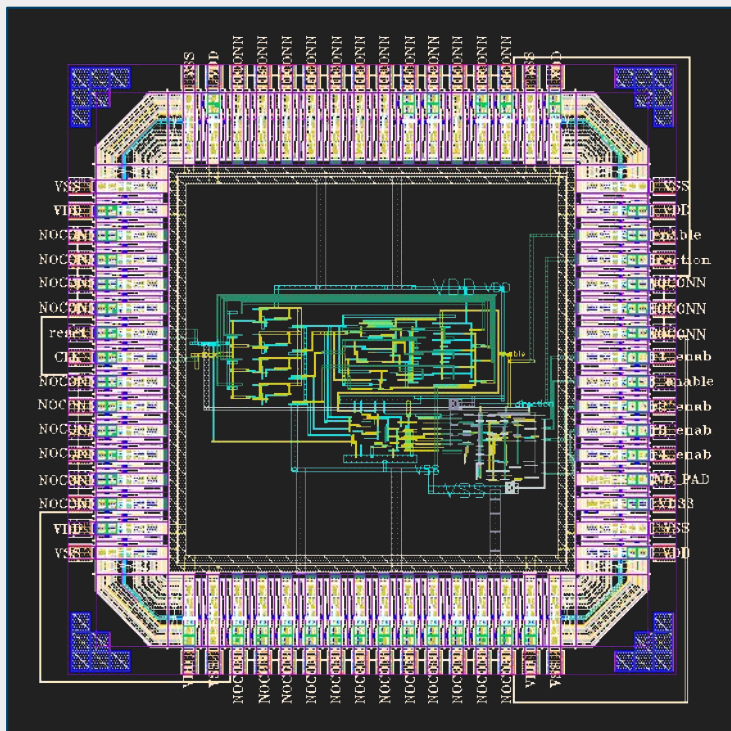
$$I_{out} = \frac{0.3}{3} = 0.1 \text{ A} = 100 \text{ mA}$$

$$R_4 = 100 \Omega, \quad R_3 = 100 \text{ k}\Omega$$



Generate 100mA current for MEMS

Top Chip Layout



Calibre - RVE v2025.3_28.17: top_chip.drc.results

File View Highlight Tools Window Setup Help

Filter: Show Unresolved top_chip, 200 Results (in 10 of 507 Checks)

Check / Cell /	Results	Flat	94	95	96	97	98	99	100
Check VIA5.S.8_Array3_4_5_M	14	896							
Check VIA5.S.7_Array2_3_4_5_M	14	896							
Check VIA5.S.6_Array1_2_3_4_5_M	14	896							
Check RES.13	64	288							
Check PO.R.3	1	1							
Check OD.R.1	1	1							
Check FPAD.R.1	9	16							
Check ESD.34g	34	562							
Check ESD.30g	48	752							
Check DRMR.1	1	1							

DRC clean **remaining issues are waivable

Calibre - RVE v2025.3_28.17: svdb top_chip

File View Highlight Tools Window Setup Help

LVS Runtime Errors

Navigator

Results

- Extraction Results
- Comparison Results

ERC

- ERC Results
- ERC Summary

Extraction Results x Comparison Results

Layout Cell / Warning Type	1
PFEED20	1

LVS clean

FSM State



Circuit Test Plan

- **Digital logic validation:**

Apply CLK, reset, enable, and direction, and verify the output timing of SS_enable and pair1_enable to pair4_enable against simulation results.

- **Block-level driver validation:**

Measure the shunt voltage, gate voltage, and output node voltage to verify the current driver operation. The output current will be estimated by $I_{out} = V_{shunt} / R_{shunt}$.

- **Chip-level validation:**

Test the full signal chain from FSM and decoder outputs to the driver interface and load response, and confirm that the correct pair is activated at the correct time.

- **Debug strategy:**

Check power rails first, then clock/reset, then digital outputs, then driver bias and feedback nodes, and finally external load and pad connections.

Conclusion

- Proposed a new actuation system to allow for **high displacements with low steady-state power consumption**
- Designed **electrothermal actuators** and **central spring mass** in accordance with simulation and calculation results for minimum desired performance
- Included **allowances** in design parameters and produced **extra dies** to ensure measurable results in MEMS testing
- Proposed a CMOS-based digital and analog control system to drive the MEMS actuator through a **state-based overlap sequence**
- Built and validated a **system-level behavioral simulation** connecting FSM logic, current generation, and MEMS displacement response
- Identified key timing, current, and displacement requirements to guide the **schematic-level circuit design** in the next stage

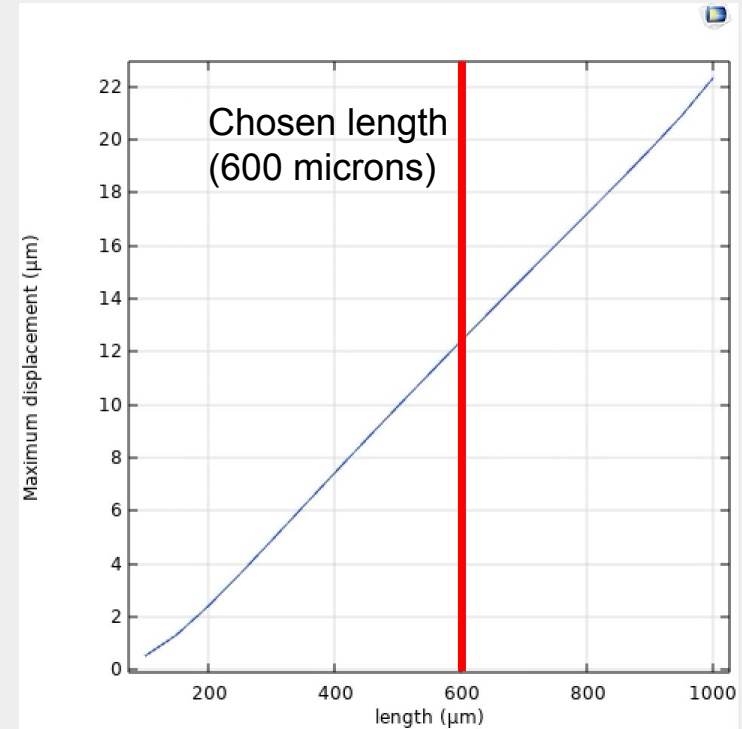


Questions?

Appendix

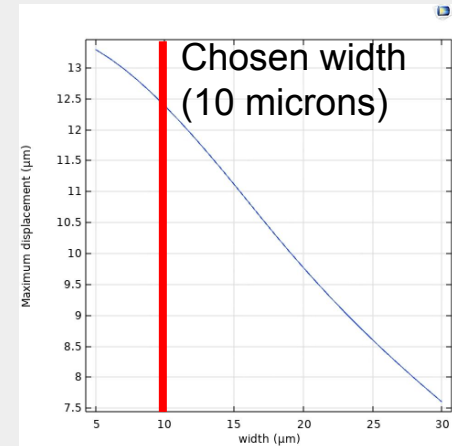
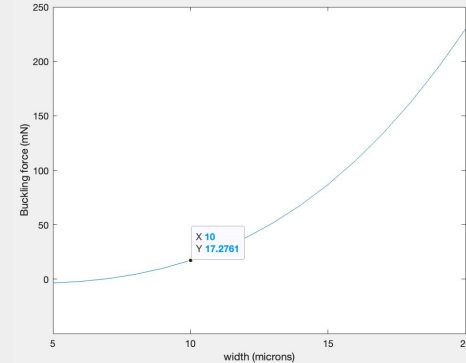
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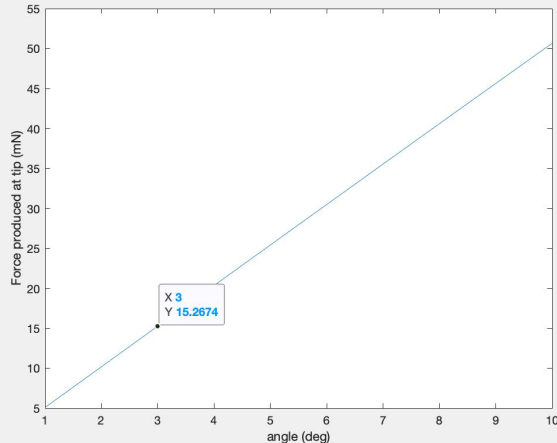
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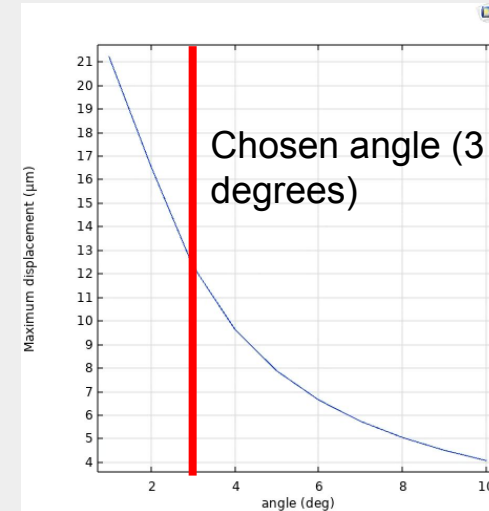


Actuator Angle

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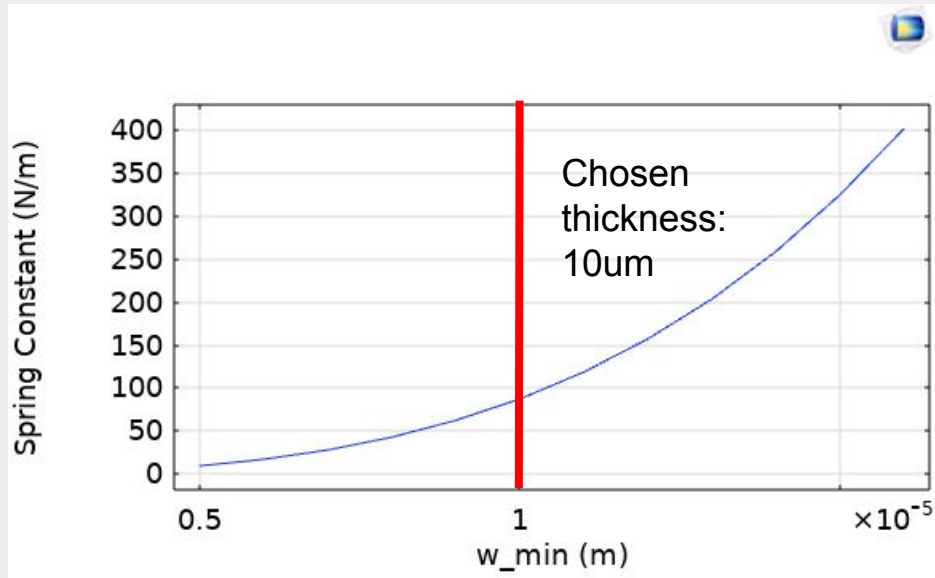


- To move the spring at the boundaries (near 50 microns) we need to produce at least 12.2 mN



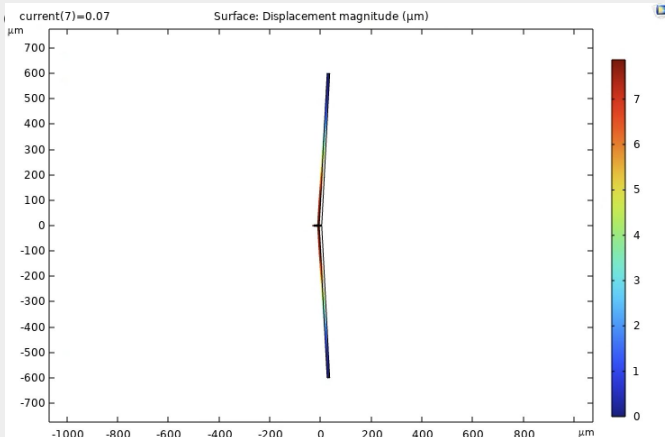
Spring Thickness

- Our spring thickness was minimized to allow for a small spring constant while fitting the 1mm design guideline
- To help with fabrication, thickness is 2 times larger than minimum design constraint and large blocks were added to help dissipate heat



Actuator Displacement

- Displacement simulation demonstrates ~8 microns of movement



- Calculation matches

F at tip of actuator:

$$\Delta T = I^2 R \cdot R_{TH}$$

$$\Delta T = I^2 R \cdot \frac{L}{k \cdot A}$$

$$330 = I^2 (60) \cdot \frac{(600 \times 10^{-6})}{(130 \cdot (100 \times 10^{-6}) \cdot (20 \times 10^{-3}))}$$

$$I = 0.0488 = \underline{48.8 \text{ mA}}$$

Displacement

600.515 mm

600 mm

93°

82°

13°

d

$$F = E \alpha \Delta T A \cdot 2 \sin \theta$$

$$E = 170 \times 10^9$$

$$\alpha = 2.6 \times 10^{-6}$$

$$\Delta T = 330$$

$$A = 100 \times 10^{-6} \times 10 \times 10^{-6}$$

$$\theta = 3^\circ$$

$$F = 15.27 \text{ mN}$$

$l_2 = l_1 (1 + \Delta T \alpha)$

$l_2 =$

trust this more than simulation

$$600.515^2 = 600^2 + d^2 - 2(600)d \cos(93^\circ)$$

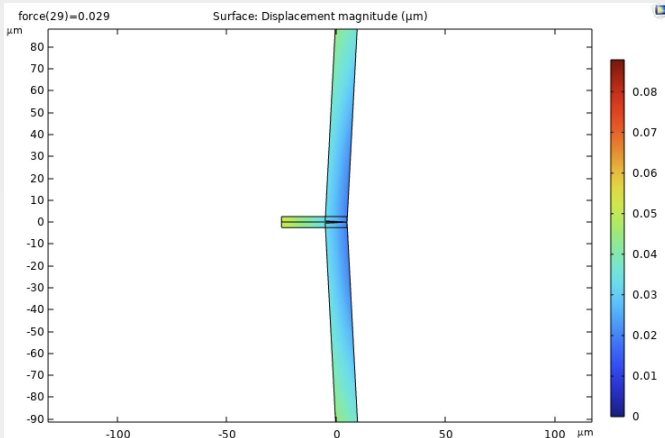
$$d^2 + 62.803d - 618.025 = 0$$

$$d = 8.649, -71.453$$

$d = 8.649 \text{ µm}$

Actuator Force

- Force simulation demonstrates equilibrium at starting position with 29 mN applied



- Calculation results in lower force, this result is assumed to be more accurate than simulation

Force at tip of actuator:

$$\Delta T = I^2 R \cdot R_{TH}$$

$$\Delta T = I^2 R \cdot \frac{L}{k \cdot A}$$

$$330 = I^2 (60) \cdot \frac{(600 \times 10^{-6})}{(130 \cdot (100 \times 10^{-6}) \cdot (20 \times 10^{-3}))}$$

$$I = 0.0488 = \underline{48.8 \text{ mA}}$$

$$F = E \alpha \Delta T A \cdot 2 \sin \theta$$

$$E = 170 \times 10^9$$

$$\alpha = 2.6 \times 10^{-6}$$

$$\Delta T = 330$$

$$A = 100 \times 10^{-6} \times 10 \times 10^{-6}$$

$$\theta = 3^\circ$$

$$F = \underline{15.27 \text{ mN}}$$

Displacement diagram showing a triangle with sides 600 mm, 600.915 mm, and 600.915 mm. The angle between the two 600 mm sides is 93°. The vertical side is labeled 'd'. The horizontal side is labeled '600 mm'. The angle between the horizontal side and the vertical side is 3°.

$$600.915^2 = 600^2 + d^2 - 2(600)d \cos(93^\circ)$$

$$d^2 + 62.803d - 618.025 = 0$$

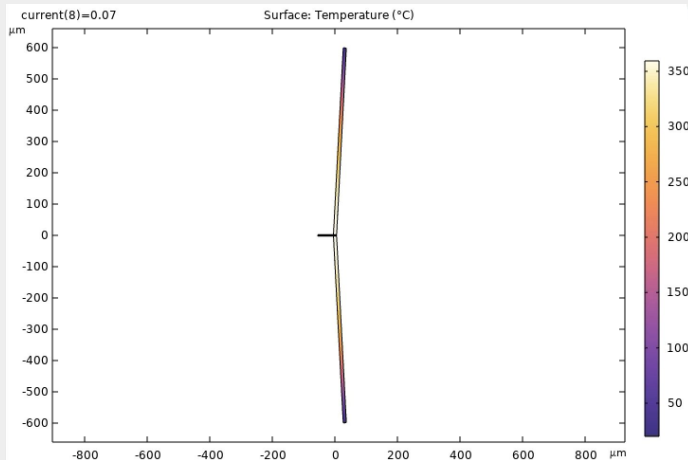
$$d = 8.649, -71.453$$

$$d = \underline{8.649 \text{ mm}}$$

Trust this more than simulation

Actuator Current

- To achieve 350° C steady state temperature, simulation shows we need 70 mA current



- Calculation is slightly lower

Handwritten calculations on a blackboard background:

Force at tip of actuator:

$$F = E \alpha \Delta T A \cdot 2 \sin \theta$$

$E = 170 \times 10^9$
 $\alpha = 2.6 \times 10^{-6}$
 $\Delta T = 330$
 $A = 100 \times 10^{-4} \times 10 \times 10^{-6}$
 $\theta = 3^\circ$

$\Delta T = I^2 R \cdot R_{TH}$
 $\Delta T = I^2 R \cdot \frac{L}{k \cdot A}$
 $330 = I^2 (60) \cdot \frac{(600 \times 10^{-6})}{(30 \cdot (100 \times 10^{-4}) \cdot (20 \times 10^{-3}))}$
 $I = 0.0488 = \underline{48.8 \text{ mA}}$

Displacement diagram showing a triangle with sides 600 mm, 600.915 mm, and 600.915 mm. The angle between the two 600 mm sides is 93°. The vertical displacement is labeled 'd'. The angle between the 600 mm side and the vertical is 13°.

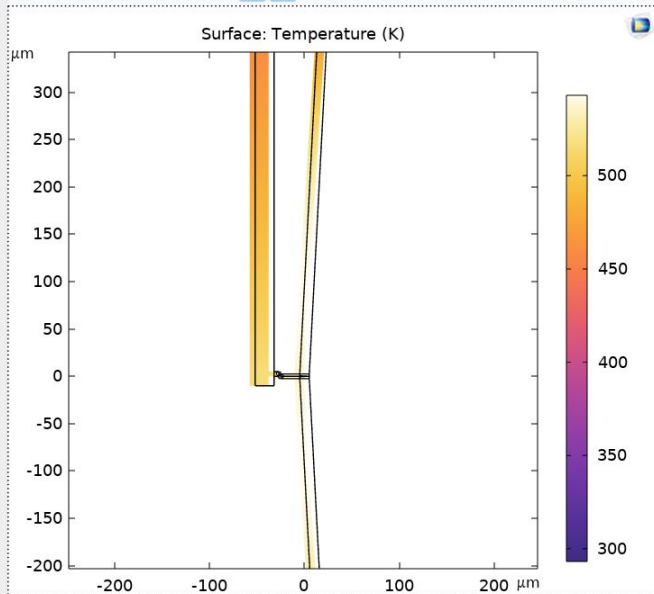
$F = 15.27 \text{ mN}$
 $l_2 = l_1 (1 + \Delta T \alpha)$
 $l_2 =$
 $600.915^2 = 600^2 + d^2 - 2(600)d \cos(93^\circ)$
 $d^2 + 62.803d - 618.025 = 0$
 $d = 8.649, -71.453$
 $d = \underline{8.649 \text{ mm}}$

Trust this more than simulation



Thermal Dispersion

Generally the contact doesn't make a large difference in the heat conducted.



Thermal dispersion by contact

$$R_{th} = \frac{L}{kA} = \frac{1400 \mu m}{(130)(1 \cdot 10^{-12} \cdot 100)} = 10,769$$

mean while R_{th} of actuator leg is 2,308

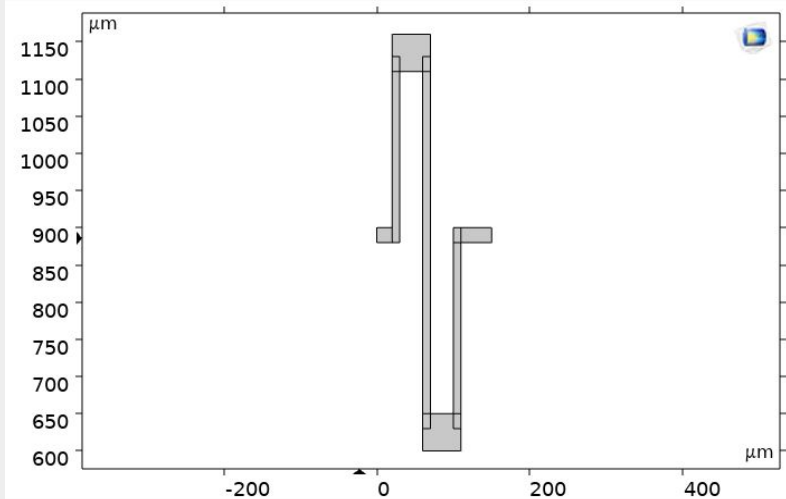
$$\text{So total } R_{th} \text{ is } (R_{actuator}^{-1} + R_{contact}^{-1})^{-1} = 1900$$

There is only a $\approx 17\%$ difference so we should be fine



Spring Constant Calculations

We got a simulated spring constant of 87.8 N/m.
The hand calculations get a similar number.



2 250 μm tall springs
1 500 μm tall spring

$$k_{\text{short}} = k_s = \frac{3EI}{L^3}$$
$$E \approx 150 \text{ GPa}$$
$$I = \frac{wt^3}{12}$$
$$k_s = \frac{E(100 \cdot 10^{-6})(10 \cdot 10^{-6})^3}{4(250 \cdot 10^{-6})^3} = 240$$
$$k_{\text{long}} = k_t = \frac{k_s}{8} = 30$$
$$k_{\text{tot}} = \left(\frac{2}{240} + \frac{1}{30} \right)^{-1} = 24 \text{ N/m}$$

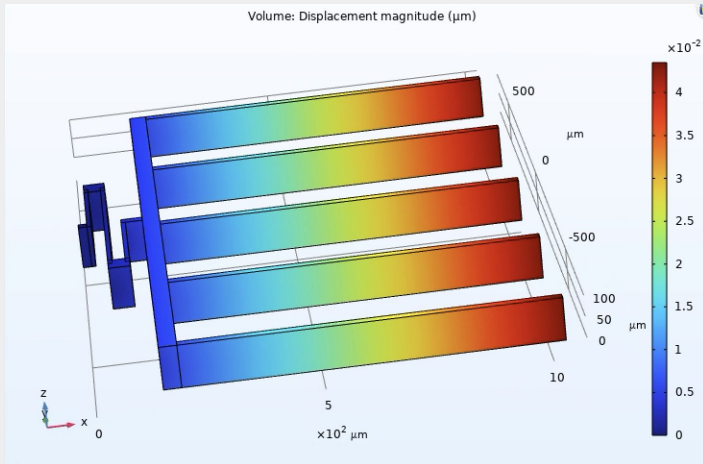
But, the springs aren't fixed-free they are more fixed-guided so

$$k = 4 \cdot 24 = 96 \text{ N/m}$$

k is somewhere in 24 N/m to 96 N/m
probably closer to 96 though

Inchworm Vertical Displacement

- Gravity isn't significant, inchworm structure will retain gap with glass beneath



- Calculation is slightly higher, but still confirms small displacement

Model as long cantilever beams

1000 mm long 10 mm thick 800 mm long 100 mm thick

Model as long cantilever beam of uniform thickness 200 mm

$$\delta_{max} = \frac{wL^4}{8EI}$$

$$\delta_{max} = 0.155 \text{ mm}$$

$$w = \rho g A = 2,329,000(9.81)(10 \times 200 \times 10^{-12})$$

$$L = 1400 \text{ mm}$$

$$E = 170 \text{ e}^9$$

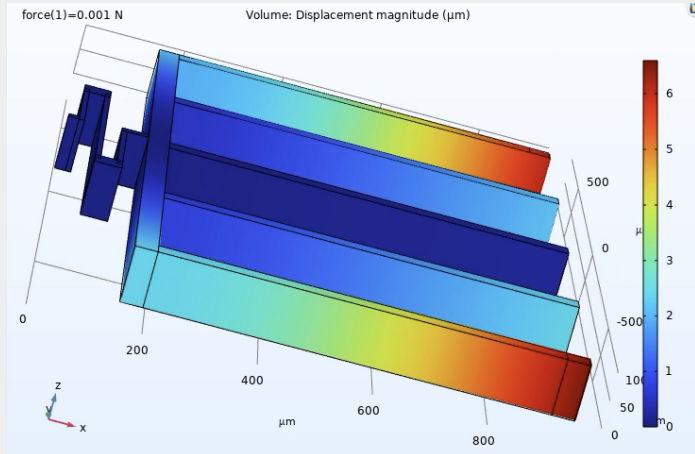
$$I = \frac{th^3}{12} = 8.333 \text{ e}^{-19}$$



Inchworm Squeeze Displacement

- Will continue to work to make this spring constant stiffer

- Calculation confirms spring constant of ~645 N/m and justifies why that value is high enough



Squeeze displacement

Force is arbitrarily large at contact point
Worst case angle of incidence 21.1°

$F_x = F \sin \theta$
 $F_y = F \cos \theta$

displacement $\rightarrow$ ensures movement to the right is stronger than movement down

$\frac{k_x dx}{k_y dy} = \frac{F_x}{F_y} = \tan \theta = 0.386$

Need $\frac{dx}{dy} > 1$ so that more displacement is in x direction than y direction

Cantilever beam $\rightarrow \frac{k_x}{k_y} = \frac{644.71}{87.8} = 7.343$ $\frac{dx}{dy} = 2.835 > 1$ ✓

our spring $\rightarrow k_y = \frac{Et^3}{4L^3} = \frac{170 \text{ GPa} (10 \text{ mm})^3 (100 \text{ mm})}{4 (750 \text{ mm})^3} = 644.741$

2 mN

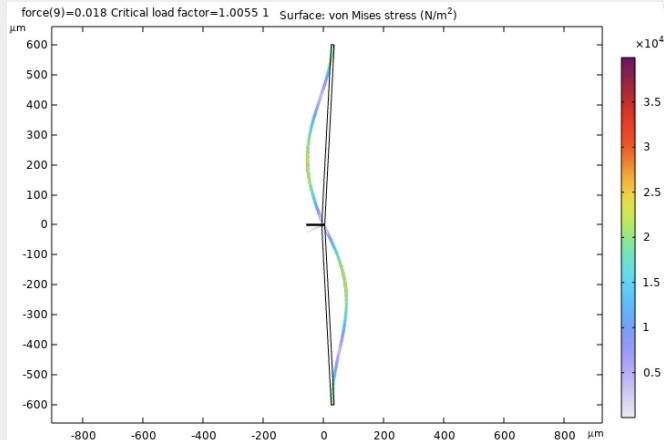
$F_N = F \cos \theta = 1.866 \text{ mN}$
 $F_F = F_N \mu = 1.866 \text{ mN} \cdot \mu$
 $F_S = F \sin \theta = 0.72$
 $0.72 = 1.866 \cdot \mu$
If $\mu \leq 0.386$ goal

$\theta = 21.1^\circ$



Buckling Forces

- Force of ~18 mN directly applied to tip of actuator to cause buckling



- Calculation verifies this result

Force for beam to buckle:

$$P_{cr} = \frac{4\pi^2 EI}{K^2 L^2}$$

$$E = 170 \text{ GPa}$$

$$I_{min} = 8.333 \text{ e-21}$$

$$L = 600 \text{ e-6}$$

$$K = \frac{1}{2}$$

$$\frac{621.4 \text{ mN}}{\text{buckle}}$$

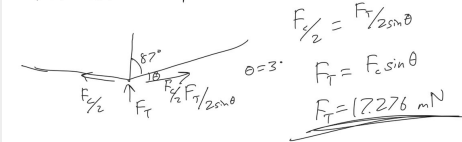
$$K = 0.5$$

$$L = 600 \text{ mm}$$

$$F = E \alpha \Delta T A \cdot 2 \cos \theta \quad (\text{compressive})$$

$$\frac{2913 \text{ mN}}{\text{produced from thermal expansion}}$$

$$\text{More additional compressive force} = 621.4 - 291.3 = 330.1 \text{ mN}$$



$$F_{buck} = \frac{F_{tip}}{2 \cos \theta} \quad (\theta = 87^\circ)$$

$$2F_{buck} = \frac{F_{tip}}{\cos \theta}$$

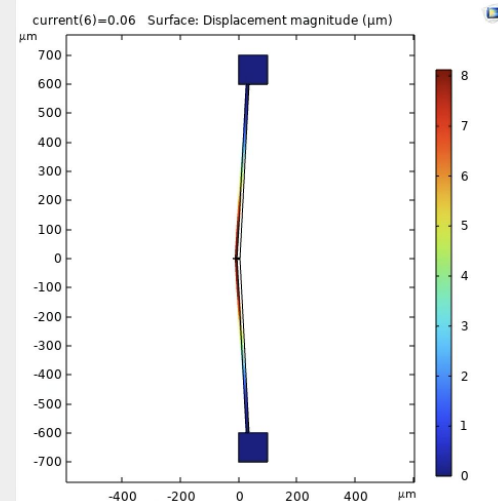
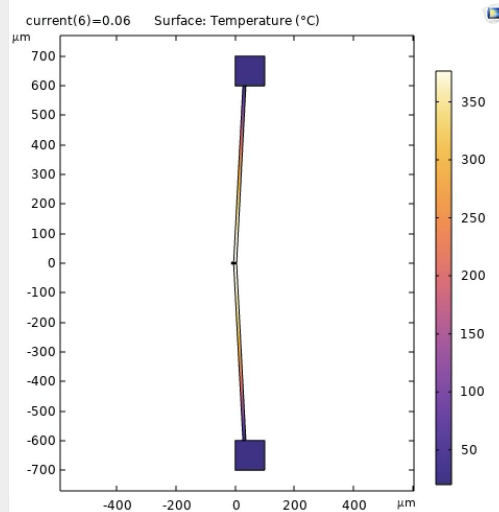
$$2F_{buck} = F_{comp}$$

$$F_{tip} = F_{comp} \cos \theta \quad (\theta = 87^\circ)$$

$$\underline{F_{tip} = 17.276 \text{ mN}} \quad \checkmark$$

Updated Simulations with TCR

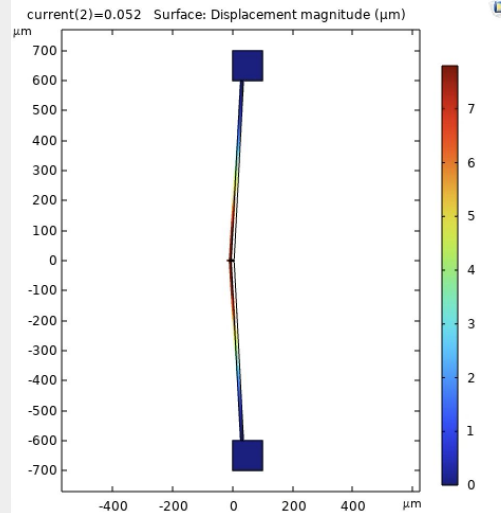
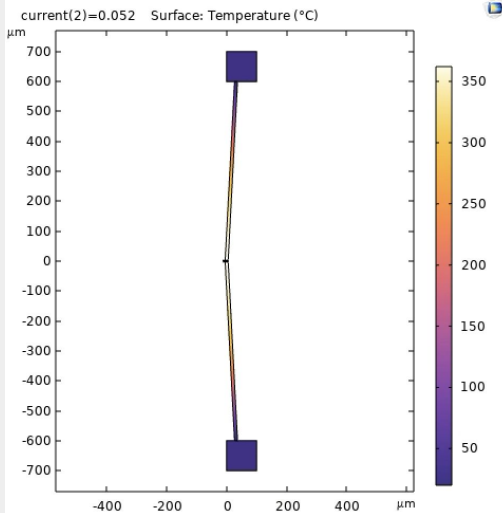
- Actuator simulations with silicon TCR indicate lower current of 60 mA is required (assuming no air associated heat loss)
- This means we get the same displacement for slightly less current





Doped Silicon Thermal Conductivity

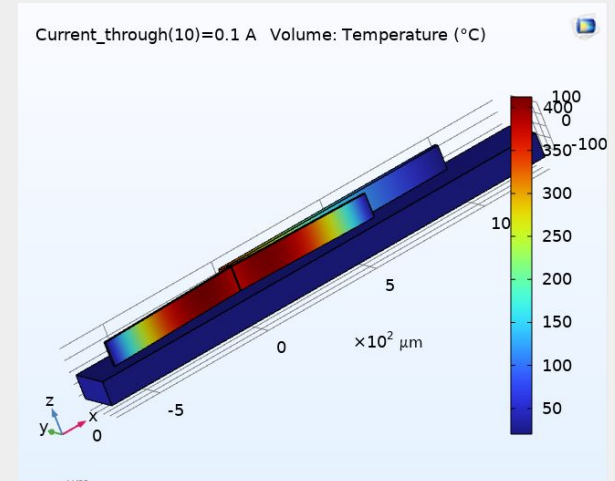
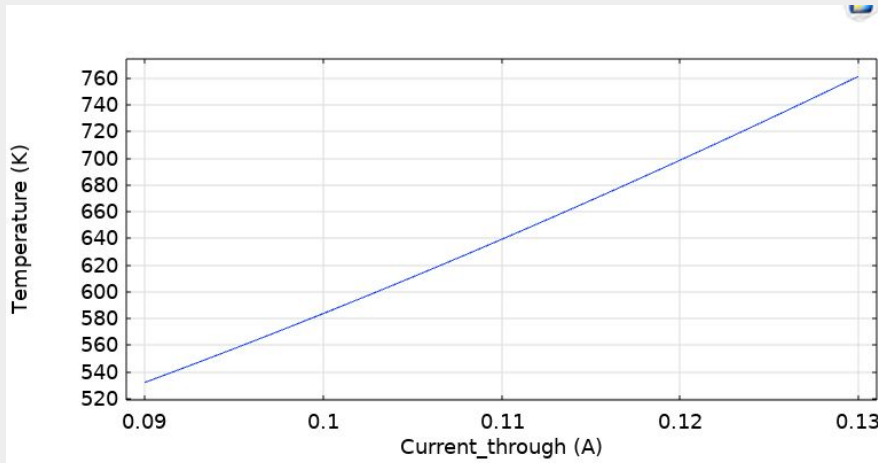
- Highly doped silicon has lower thermal conductivity ($\sim 100 \text{ W/m}\cdot\text{K}$), so current requirement may be even lower ($\sim 52 \text{ mA}$)



- In general, we are slightly overspec'ing our design (designing for 60-100 mA) in case some results are not as expected, especially with air-related heat loss

Air Thermal Simulations

We simulated the actuator with heat loss through air and with heat loss through contact to make sure that the arm stays extended. We did neglect radiation but some hand calcs showed it had a minimal effect.



Thermal time constant

Thermal time constant:

$$\tau = C_{th} R_{th} \quad m = 1200 \cdot 15 \cdot 1000 \text{ p} \cdot 10^{-18}$$

$$C_{th} = m c_p \approx m 700 \approx 3 \text{ pJ/K}$$

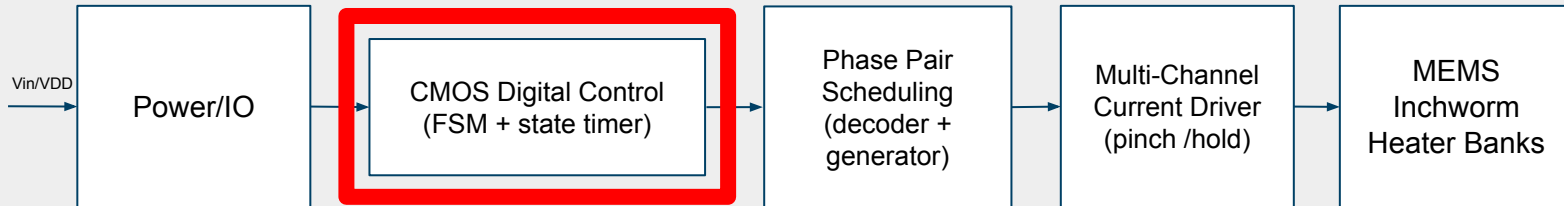
$$R_{th} = \frac{1}{2} \left(\frac{600 \text{ nm}}{100 (15 \text{ nm} - 100 \text{ nm})} \right) \approx 2000$$

$$\tau \approx 6$$

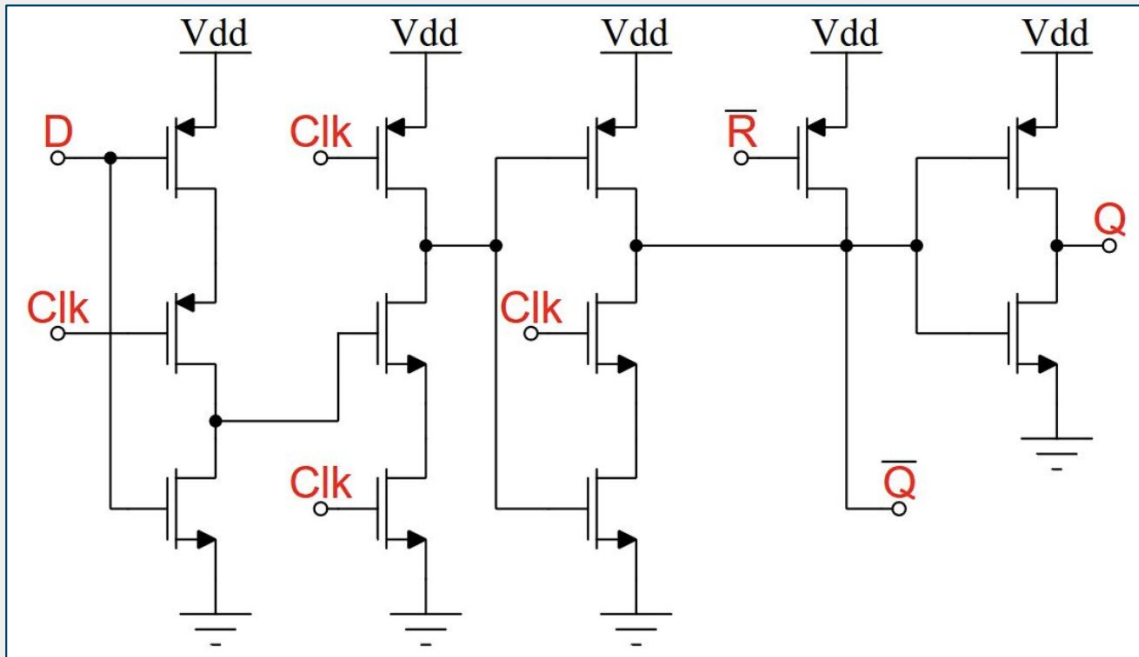
CMOS Digital Control

The brain of the system that produces which stage is active/how long that stage stays active

- FSM state storage: implemented with D flip-flops (clocked counter)



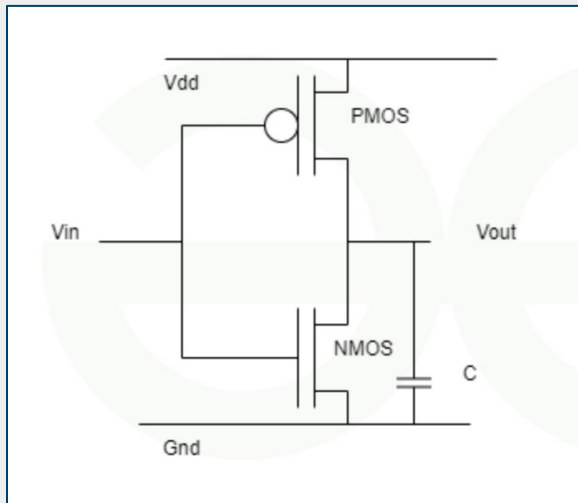
FSM logic



D Flip Flop:

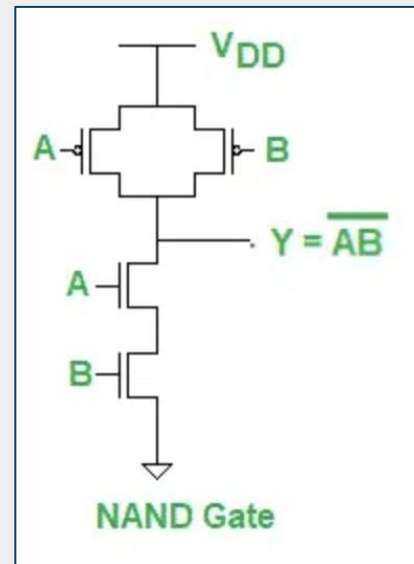
Used to store the current state and counter values, enabling the FSM to update its state synchronously on each clock edge

FSM logic



Inverter:

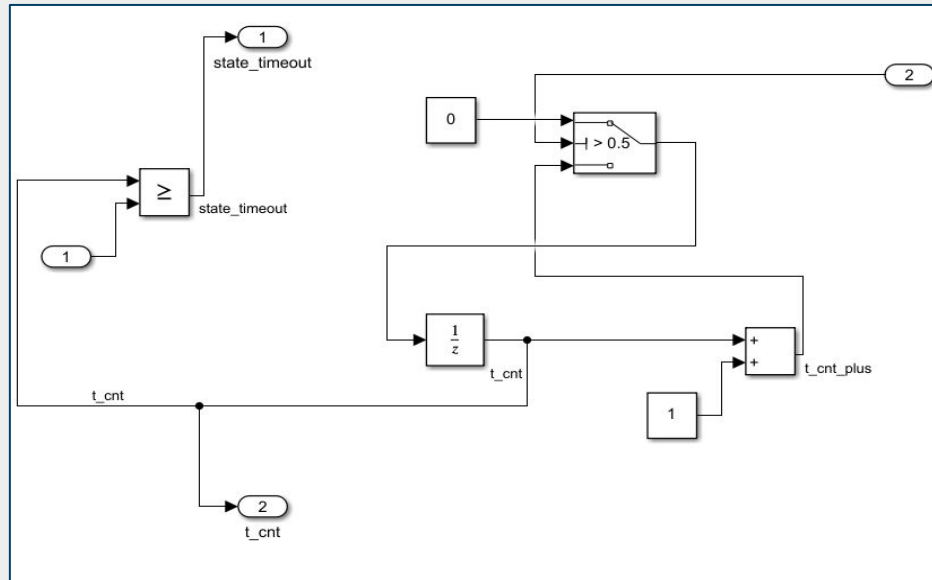
Used for phase inversion and decoder logic



NAND:

Used for pair decoding and enable signals

State Timer



State Timer Simulink Model

- Implements a counter-based timer to track how long the system stays in a state
- Switch resets timer when timeout occurs
- Outputs:
 - t_cnt = current state time counter
 - state_timeout = timeout flag



State Timer

State Duration Tracking: Acts as a discrete counter to measure exactly how long the system remains in its current state.

Asynchronous Reset: Instantly resets the internal counter to zero whenever an FSM_change pulse is detected.

Timeout Trigger: Drives the state_timeout output high (5V) the moment the active count reaches the target threshold (T_state).

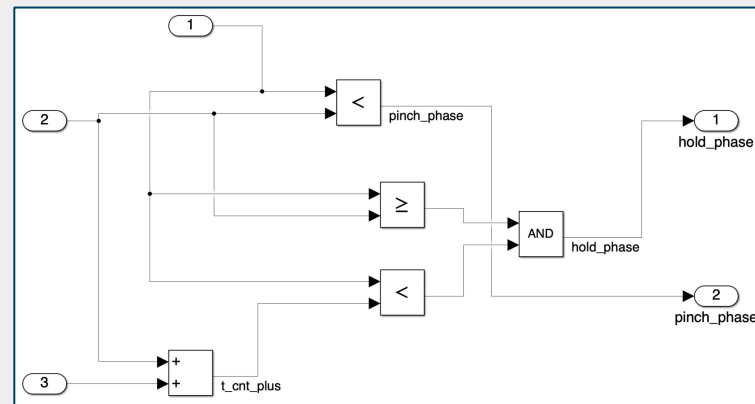
Configurable Design: Uses built-in parameters for the target limit and clock speed, replacing hardwired constant blocks for easy tuning.

```
1 'include "constants.vams"
2 'include "disciplines.vams"
3
4 module daStateTimer(
5     input in_FSM_change,
6     output out_state_timeout,
7     output out_t_cnt
8 );
9
10 electrical in_FSM_change, out_state_timeout, out_t_cnt;
11 parameter integer T_state = 60;
12 parameter real sample_time = 1n;
13 parameter real Vhigh = 5.0;
14 parameter real Vlow = 0.0;
15 parameter real t_rise = 5n;
16 parameter real t_fall = 5n;
17
18 integer internal_count;
19 real timeout_logic;
20
21 analog begin
22
23     @(initial_step) begin
24         internal_count = 0;
25     end
26
27     @(timer(0, sample_time)) begin
28         if (V(in_FSM_change) > 0.5) begin
29             internal_count = 0;
30         end else begin
31             internal_count = internal_count + 1;
32         end
33     end
34
35     if (internal_count >= T_state) begin
36         timeout_logic = Vhigh;
37     end else begin
38         timeout_logic = Vlow;
39     end
40
41     V(out_state_timeout) <+ transition(timeout_logic, 0, t_rise, t_fall);
42     V(out_t_cnt) <+ transition(internal_count, 0, t_rise, t_fall);
43 end
44 endmodule
45
```

Phase Pair Scheduling

Works as a translator between high and low level signals VDD_digital: FSM, counters, DFFs, decoders

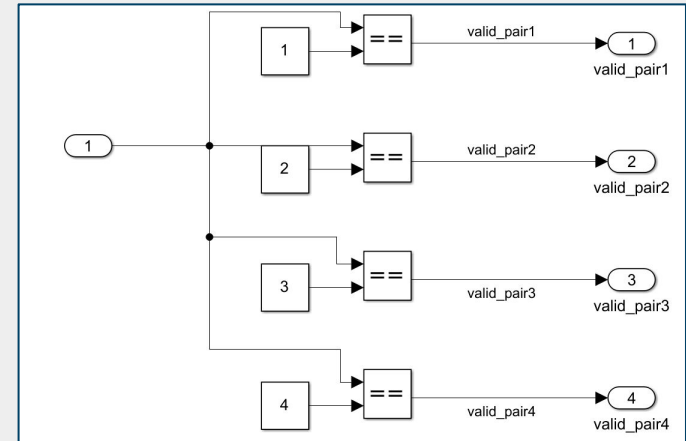
- Phase generator: takes FSM state and decides:
 - Which pair is “active” for the inchworm
 - Whether system is in *pinch* or *hold*



Phase Generator Simulink Model

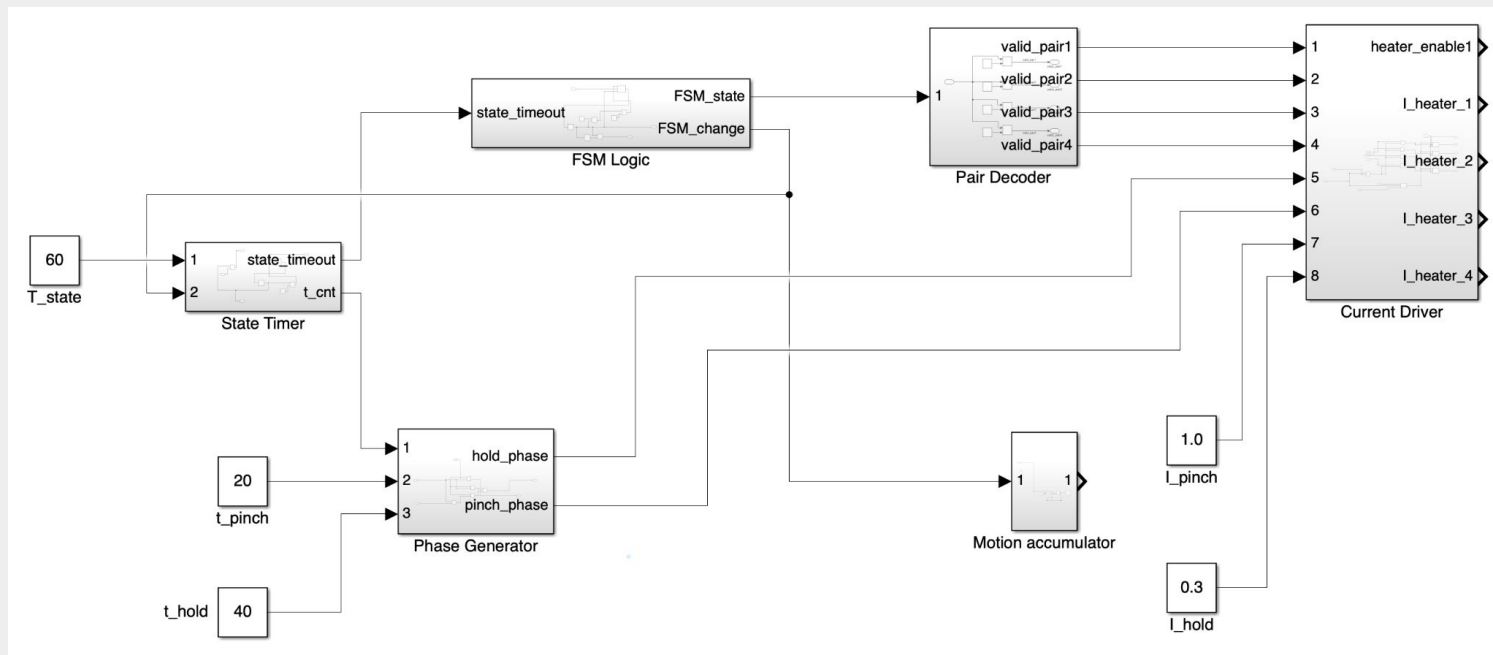
Phase Pair Scheduling

- Pair decoder: takes a pair ID and produces valid_pair1, valid_pair2, valid_pair3, or valid_pair4 based on equality comparators



Pair Decoder Simulink Model

Circuit Simulink Block Diagram





Pair Decoder

- Converts analog FSM state voltage ($\approx 1V$, $2V$, $3V$, or $4V$) into one-hot enable signals for 4 heater/actuator pairs
- Input (sel_in) carries the state code as a voltage.
- Initial Step forces all voltages down to zero. (temporary)
- Chose $5V$ based on actuator specific power

```
1 'include "constants.vams"
2 'include "disciplines.vams"
3
4 module daPairDecoder (
5     inout sel_in,
6     output valid_pair1,
7     output valid_pair2,
8     output valid_pair3,
9     output valid_pair4
10 );
11     electrical sel_in, valid_pair1, valid_pair2, valid_pair3, valid_pair4;
12
13     parameter real Vhigh = 5.0;
14     parameter real Vlow = 0.0;
15     parameter real Vtol = 0.2;
16     parameter real t_rise = 5n;
17     parameter real t_fall = 5n;
18
19     real vin;
20
21     analog begin
22         @(initial_step) begin
23             V(valid_pair1) <+ Vlow;
24             V(valid_pair2) <+ Vlow;
25             V(valid_pair3) <+ Vlow;
26             V(valid_pair4) <+ Vlow;
27         end
28         vin = V(sel_in);
29         V(valid_pair1) <+ transition( abs(vin - 1.0) <= Vtol ? Vhigh : Vlow, 0, t_rise, t_fall);
30         V(valid_pair2) <+ transition( abs(vin - 2.0) <= Vtol ? Vhigh : Vlow, 0, t_rise, t_fall);
31         V(valid_pair3) <+ transition( abs(vin - 3.0) <= Vtol ? Vhigh : Vlow, 0, t_rise, t_fall);
32         V(valid_pair4) <+ transition( abs(vin - 4.0) <= Vtol ? Vhigh : Vlow, 0, t_rise, t_fall);
33     end
34
35 endmodule
```

Current Driver Cadence Schematic

Input Differential Pair (MN1–MN2)

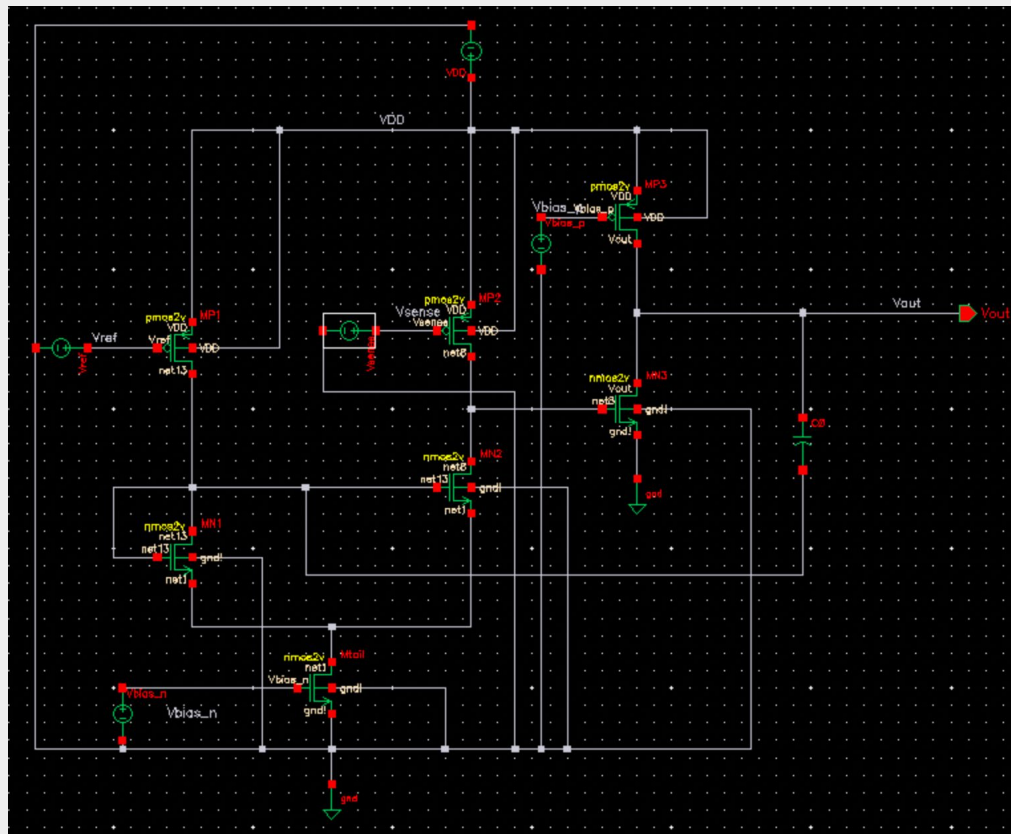
Active Load / Current Mirror (MP1–MP2)

Tail Current Source (Mtail)

Second Gain Stage (MN3 / MP3)

Bias Network (Vbias_p / Vbias_n devices)

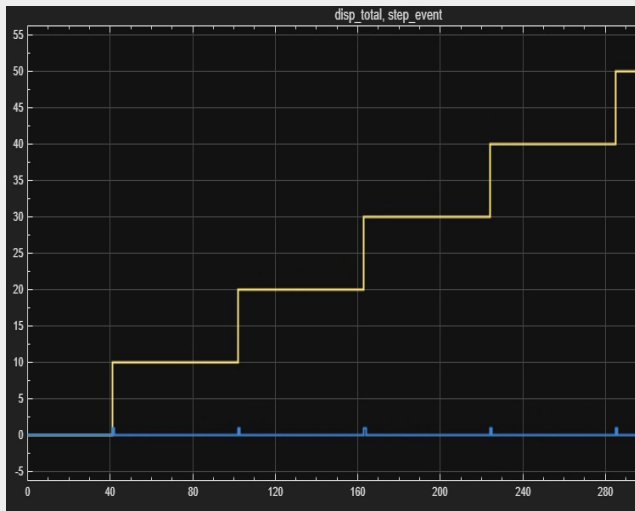
Output Node (Vout)



Worst Case Scenario

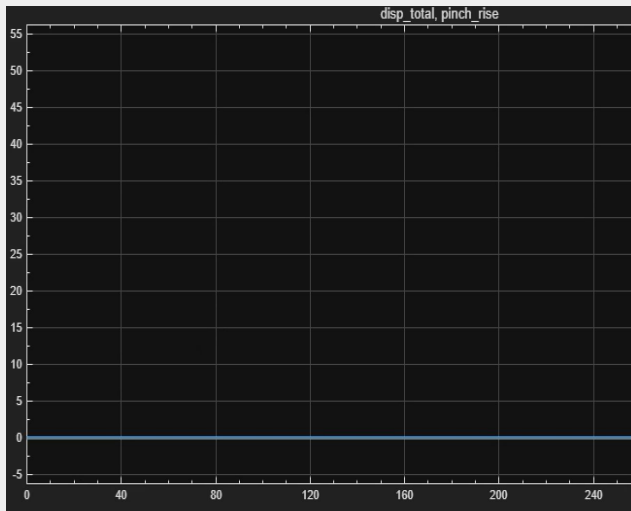
Case 1: Actuation Efficiency Degradation

Reduced displacement per pinch step lowers the final reachable displacement.



Case 2: Current Driver Degradation

Insufficient heater current may weaken or suppress valid displacement steps



Case 3: Timing Degradation

A shortened pinch window reduces timing margin for effective actuation

Specification of Circuit Requirement

Circuit Block	Function	Key specs	Implemented by
State Timer	State duration control	CLK Frequency = 33.33 Hz	CMOS digital
FSM logic	Sequential state control	10-state cyclic sequence $S0 \rightarrow S1 \rightarrow \dots \rightarrow S9 \rightarrow S0$ one update per timeout reset to S0	CMOS digital
Pair Decoder	State-based actuation decoding	Generates SS_enable and pair1~4_enable state-dependent overlap control	CMOS digital

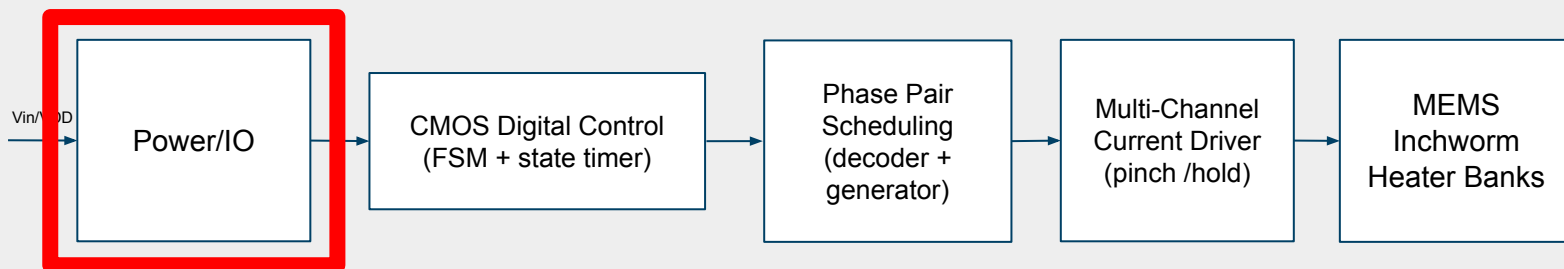
Specification of Circuit Requirement

Circuit Block	Function	Key specs	Implemented by
Phase Generator	Safe-state and pair overlap sequencing	S0: no current overlap sequence includes SS+P1, SS+P1+P2, SS+P2, SS+P2+P3, SS+P3, SS+P3+P4, SS+P4, P4	CMOS digital
Current Driver	Heater current control	$I_{\text{act}} = 80 \text{ mA}$	CMOS analog
MEMS	Displacement response	13 μm for the first cycle 16 μm for the later cycles thermal time constant: $\sim 7\text{ms}$ thermal capacitance: $\sim 4\mu\text{J/K}$	CMOS digital

Power Management

Delivers stable rails to the three main electrical loads

- VDD_digital: FSM, counters, DFFs, decoders
- VDD_driver: current drivers that push 60-100 mA currents into heater resistors



References:

1. Khazaai, J., Khir, M. H. M., Qu, H., & Slicker, J. (2010). Design and fabrication of a low power electro-thermal V-shape actuator with large displacement. In Proc. NSTI-Nanotech (Vol. 2, pp. 681-684).
2. Pal, J., Zhu, Y., Lu, J., & Dao, D. V. (2013, June). A novel electrothermally actuated RF MEMS switch for wireless applications. In 2013 IEEE 8th Conference on Industrial Electronics and Applications (ICIEA) (pp. 1594-1598). IEEE.